

Overreaction in Household Inflation Expectations: Evidence from China*

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Abstract

Using linked household data from seven waves of the China Family Panel Studies, I show that upward revisions in inflation expectations predict subsequent forecast errors in the opposite direction ($\hat{\beta} = -0.55$, $N \approx 13,000$). The pattern is robust across demographic subgroups, alternative expectation codings, and progressive controls with province $\times$ wave fixed effects. Aggregate depositor-survey data display the same sign. A joint specification nesting revision-driven overreaction and lagged-error persistence reveals error reversals rather than sluggish updating, though the short aggregate sample limits sharp discrimination between models. The data display a reversal pattern more consistent with overreaction than with a pure sluggish-updating account, though the paper does not claim full structural identification of belief formation.

JEL Codes: D84, E31, E71, G53

Keywords: inflation expectations, overreaction, household surveys, diagnostic expectations, China

*All errors are my own. Replication code and data documentation are available at https://github.com/Manjusaka666/WP\protect_BFSU.

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1 Introduction

Households in China overreact to inflation signals. Using seven waves of the China Family Panel Studies (CFPS, 2010–2022), I regress each household’s forecast error on its prior expectation revision and estimate $\beta = -0.55$ ($p < 0.001$, $N \approx 13,000$) with province $\times$ wave fixed effects (Table 10). The sign is unambiguous: a one-point upward revision in expected inflation predicts a 0.55-point more negative forecast error at the next interview. This pattern does not come from one narrow subgroup. It appears across income quartiles, education levels, age bins, and the urban-rural divide, so the wedge between expected and realized inflation is broad rather than idiosyncratic.

China is a useful setting for studying household expectation formation. Food prices, especially pork, are salient to consumers and volatile enough to generate large belief revisions; the 2018–2019 African Swine Fever episode is one recent example. At the same time, monetary policy communication from the People’s Bank of China remains less transparent than in major inflation-targeting central banks, so households receive strong price signals but weak anchoring guidance. This combination makes both overreaction and slow updating plausible ex ante, which means the data can discriminate between them rather than merely confirm a prior.

I make three contributions. First, I establish the main fact at the household level: individuals who revise inflation expectations upward subsequently make more negative forecast errors, consistent with overshooting. This holds across demographic and geographic cells in the CFPS panel with stable magnitude after alternative survey weights and multiple coding choices for qualitative responses. Second, I provide aggregate corroboration using quarterly PBoC depositor-survey data, which yield $\beta = -2.24$ ($p = 0.055$, $N = 32$). The sign matches the household-level result, and the timing aligns with quarters of salient food-price news. Third, I estimate a joint specification that nests both overreaction and persistence channels to ask which explains forecast errors conditional on the other. Error reversals remain present after controlling for persistence, more consistent with overreaction than with pure sluggish updating. A supplementary test using the China Household Finance Survey (CHFS) asks whether persistent differences in media exposure alone explain belief intensity in cross section. The estimated media-channel slope is $\beta = -0.005$ ($SE = 0.022$), economically negligible and statistically indistinguishable

from zero, which suggests that overreaction is event-driven rather than a fixed function of information access.

I state the limits of this paper explicitly. I do not claim point identification of a structural diagnostic-expectations parameter. I do not claim a decisive test against all information-rigidity models. I do not claim a policy-invariant welfare mapping from overreaction to output losses. The aggregate sample is short ($N = 32$), so macro regressions confirm sign and timing but cannot pin down precise elasticities. The CFPS expectation variable is biennial and qualitative, which compresses adjustment dynamics and limits high-frequency tracing. The salience IV has weak first-stage power ($F = 1.16$, Table 8, Panel A), so I treat IV estimates as directional checks, not as a basis for sharp causal claims. The CHFS test is a single cross-section without panel leverage. These constraints narrow interpretation but do not overturn the core household-level fact: a large, negative, and precisely estimated revision-error elasticity that replicates across subsamples.

Section 2 presents the conceptual framework. Section 3 describes the data, and Section 4 presents the identification strategy. Section 5 reports baseline results. Section 6 examines dynamic interpretation. Section 7 reports heterogeneity. Section 8 collects robustness checks. Section 9 concludes. A forecasting application appears in Appendix G.

2 Conceptual Framework

Two frictions can distort household inflation expectations: diagnostic overreaction and information rigidity. Diagnostic expectations follow representativeness. Agents overweight states made vivid by recent news and underweight base rates (Kahneman and Tversky; Gennaioli and Shleifer). In macro applications, revisions overshoot and then partially reverse (Bordalo, Gennaioli, and Shleifer; Bordalo et al.). Information rigidity imposes a different friction. Households update slowly because acquiring and processing information is costly (Sims; Coibion and Gorodnichenko). Errors decay gradually rather than reversing.

I estimate both channels jointly and use sign restrictions to separate them. The question is which channel accounts for more forecast-error variation in a setting with high food-price salience and relatively opaque policy communication.

2.1 Diagnostic vs rigidity predictions

Define the forecast error and forecast revision as

$$FE_{t+1} \equiv \pi_{t+1} - \mu_t, \quad FR_t \equiv \mu_t - \mu_{t-1},$$

where π_{t+1} is realized inflation and μ_t is the expectation formed at time t . I estimate a nested specification:

$$FE_{t+1} = \alpha + \beta FR_t + \lambda FE_t + X_t' \Gamma + u_{t+1}.$$

The coefficient β captures revision overshoot. The coefficient λ captures persistence in inherited forecast gaps.

Under diagnostic expectations, upward revisions are too large relative to subsequent realizations. A positive revision is followed by a negative future error:

$$\beta < 0 \iff \mathbb{E}[FE_{t+1} \mid FR_t > 0] < 0. \quad (1)$$

This is the overreaction moment in [Bordalo et al.](#) It implies an overshoot-and-correction profile.

Under information rigidity, households close information gaps slowly. Errors persist because updating is incomplete:

$$\lambda > 0 \iff \mathbb{E}[FE_{t+1} \mid FE_t > 0] > 0. \quad (2)$$

This is the sluggish-updating moment in [Coibion and Gorodnichenko](#). Adjustment is monotone rather than reversal-based.

The two channels can coexist. A household may underreact to low-salience background information and overreact when a salient shock dominates attention. In that case, $\beta < 0$ and $\lambda > 0$ can hold in the same sample. Joint estimation is therefore necessary because separate regressions can misattribute one channel to the other.

2.2 Chinese institutional context

China provides conditions under which both channels can be strong. On the salience side, food inflation is visible in daily shopping, and pork prices are the primary amplifier of household inflation narratives. African Swine Fever episodes created abrupt supply losses and large price spikes. When households observe these spikes repeatedly, recent food news becomes a dominant signal for aggregate inflation. This pushes behavior toward representativeness.

On the rigidity side, households face a communication environment with less explicit forward guidance and less transparent policy mapping than benchmark inflation-targeting regimes (Chen and Zha). When households cannot map policy messages into a clear inflation path, they update with delay. That delay raises forecast-error persistence, the signature captured by $\lambda > 0$.

Policy implications differ by mechanism. If diagnostic overreaction dominates, communication should counter salience cascades and anchor expectations around medium-run fundamentals. If rigidity dominates, policy should improve frequency, clarity, and predictability of communication to reduce the stock of uninformed households. Both channels favor better communication, but the optimal design differs.

2.3 Testable predictions summary

The design evaluates four empirical moments the data can speak to. First, the revision-error sign. Diagnostic models predict $\beta < 0$: larger upward revisions should precede more negative forecast errors (Bordalo et al.). Pure rigidity does not require this overshoot moment.

Second, the lagged-error sign. Rigidity predicts $\lambda > 0$: positive errors today should carry into next-period errors (Coibion and Gorodnichenko). Diagnostic models allow weak or negative lagged-error dependence when correction follows overshoot.

Third, cross-sectional media patterns. Diagnostic models predict stronger overreaction where inflation news is more salient. Rigidity models predict faster updating where information access is better, implying weaker persistence in those groups.

Fourth, dynamic patterns. Diagnostic dynamics are front-loaded: revision shocks

move errors quickly, then attenuate or reverse. Rigidity dynamics are gradual: errors decay slowly with same-sign persistence across adjacent horizons. These profiles provide a second discriminating margin beyond static sign restrictions.

I do not claim these moments deliver a decisive structural test between the two models. The paper documents patterns consistent with overreaction and quantifies how much each channel contributes to forecast-error variation in Chinese household data.

Table 1 positions the present paper relative to the closest work in the household-expectations literature. The key distinction is data type and identification source: this paper uses a nationally representative household panel with province $\times$ wave fixed effects rather than professional forecaster data, randomized information treatments, or price-scanner exposure. The mechanism is inferred from reduced-form moments rather than directly measured.

Table 1: Literature Positioning: Comparison of Empirical Designs

Paper	Data	Horizon	Main Expectation Object	Identification	
Bordalo et al.	SPF professional forecasts	Quarterly	Revision $\rightarrow$ error predictability	Cross-forecaster variation	N
Coibion and Gorodnichenko	SPF, Michigan, Livingston	Quarterly	Revision $\rightarrow$ error dynamics	Time-series moments	N
Cavallo, Cruces, and Perez-Truglia	Online survey experiment	3–6 months	Belief updating after info treatment	Randomized information	Y
D’Acunto et al.	German household panel	Monthly	Grocery exposure $\rightarrow$ expectations	Price scanner data	Y
Malmendier and Nagel	Michigan Survey	Quarterly	Lifetime experience $\rightarrow$ expectations	Cohort variation	N
Andre et al. (2022)	Survey experiment	N/A	Subjective macro models	Randomized scenarios	Y
This paper	CFPS + PBoC	Biennial / Quarterly	Revision $\rightarrow$ error (household)	Province $\times$ wave FE	N

Notes: This table compares the empirical design of the present paper with related work on household expectation formation. “Mechanism Direct?” indicates whether the paper hypothesized belief-formation channel or infers it from reduced-form moments.

3 Institutional Setting and Data

This section describes the three survey datasets and the quarterly macro controls used in the main specifications. The PBoC depositor survey provides the aggregate quarterly test. The CFPS household panel provides a within-household test at biennial frequency. The CHFS contributes a single cross-section used for one mechanism boundary condition.

3.1 PBoC Depositor Survey

The People’s Bank of China Urban Depositor Survey provides the aggregate expectation series at quarterly frequency. The working sample spans 2011Q1–2025Q3. After merging with macro controls and requiring non-missing revisions and forecast errors, the baseline aggregate regressions use $N = 32$ quarterly observations (Table 2).

The survey is fielded quarterly in major cities and reports aggregate qualitative expectation shares. The estimation sample is shorter than the raw archive because the baseline regressions require complete macro controls and consistent timing for revisions and next-quarter forecast errors.

The expectation question is qualitative and published on a five-point response scale. I convert the published response shares into a quantitative expectation using a Carlson–Parkin mapping, with full construction details in Appendix C and Appendix E. The quarterly aggregate revision and error are

$$FR_t = \mu_t - \mu_{t-1}, \quad FE_{t+1} = \pi_{t+1}^{QoQ} - \mu_t.$$

This timing aligns forecast formation at quarter t with realized inflation in $t + 1$. Alternative allocations for uncertain responses are retained for sensitivity analysis; their implications for quantified expectations are summarized later in Table 6.

Figure 1 plots the aggregate revision and next-quarter error series used in the baseline time-series test.

Table 2: Sample Overview

	Value
<i>Panel A: Sample Coverage</i>	
Main sample start	2013Q4
Main sample end	2022Q3
Main sample observations	32
Total available quarters	48
<i>Panel B: Data Availability</i>	
Alternative expectations measure available	Yes

Notes: The main sample requires non-missing forecast error, forecast revision, uncertainty measures, and macro controls. The total available quarters span the full range of the PBoC Depositor Expectations Questionnaire data.

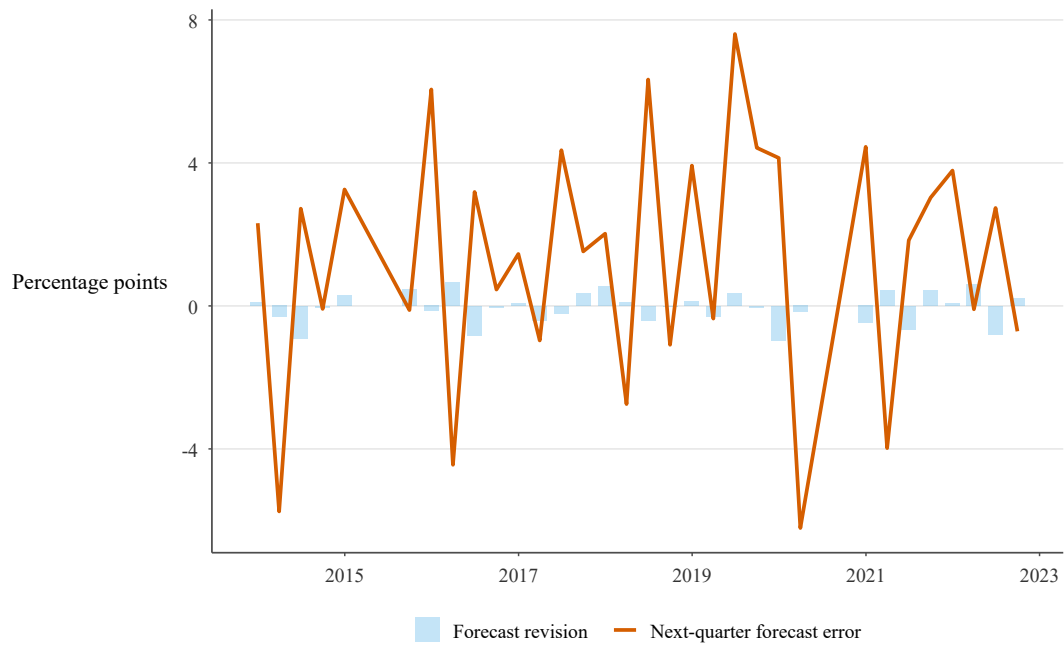


Figure 1: Aggregate forecast revision (bars) and subsequent one-quarter-ahead forecast error (line), 2011Q1–2025Q3.

3.2 CFPS Household Panel

The China Family Panel Studies (CFPS) is a nationally representative household panel administered by the Institute of Social Science Survey at Peking University (Xie and Hu; Xie, Hu, and Zhang). I use seven waves (2010, 2012, 2014, 2016, 2018, 2020, 2022). The analysis sample contains about 13,000 household-wave observations after requiring non-missing expectation responses, province identifiers, and panel linkage.

The CFPS cross-year identifier links respondents across waves and allows construction of within-household changes in inflation beliefs. Household-level covariates include age, education, hukou status, and income. The panel is unbalanced, so all specifications are estimated on the available linked observations for each wave pair.

Measurement. The inflation-expectation item is a three-point qualitative question: prices rise, unchanged, or fall. I code $\mu_{it} \in \{1, 0, -1\}$ for household i at wave t . For households observed in consecutive waves, the revision is

$$\Delta\mu_{it} = \mu_{it} - \mu_{i,t-1}.$$

Forecast errors are constructed by matching each household to province-level realized CPI inflation between t and $t + 1$, then comparing outcomes with the direction implied by μ_{it} .

Three features of this measure deserve explicit comment. First, the question is coarse. A three-point scale cannot capture the magnitude of expected price changes, only direction. Second, the survey is biennial, so revision horizons are two years rather than one quarter. Third, the forecast-error object is indirect: I match a qualitative household response to later realized provincial CPI, which is a continuous variable measured at a different level of aggregation.

Why the exercise is still informative. Despite these limitations, the CFPS test has four properties that make it useful. The three-point coding is consistent across all seven waves, so measurement error in the dependent variable is stable over time. The time ordering is clean: revisions use information available at t , while errors use realizations after t . The specification includes province $\times$ wave fixed effects, so the identifying varia-

tion is within-cell heterogeneity in household belief revisions. This absorbs any aggregate provincial inflation trend that might mechanically link revisions and errors. Finally, the estimated coefficient is stable across alternative control sets and demographic subgroups (Section 8).

Table 3 reports wave-by-wave CFPS coverage and key composition statistics used in the panel construction.

Table 3: CFPS Panel Summary Statistics by Wave

	(1) 2012	(2) 2014	(3) 2016	(4) 2018	(5) 2020	(6) 2022
<i>N</i>	20,907	23,673	22,327	26,327	21,222	20,614
<i>Price expectations (%)</i>						
Expecting price rise	16.1	21.5	20.5	21.6	22.7	22.4
Expecting stable	32.1	30.5	27.8	23.9	23.6	22.4
Expecting price fall	51.8	48.0	51.7	54.5	53.7	55.2
<i>Demographics</i>						
Mean education (years)	1.6	8.5	8.9	8.5	7.9	6.8
Urban hukou (%)	72.7	73.1	—	73.2	81.3	82.0
Mean age	39.2	40.2	40.2	—	—	—
Mean income (CNY)	13,777	11,340	7,839	16,152	21,125	26,120

Notes: Price expectations are coded as 1 (rise), 0 (stable), -1 (fall). Percentages computed over non-missing observations within each wave. Income is in nominal CNY. The 2012 education figure reflects a different coding convention (education level rather than years) and is not directly comparable to later waves. “—” indicates variable unavailable for that wave.

Horizon mismatch between CFPS and PBoC. The PBoC survey is quarterly and the CFPS is biennial, so the two datasets test the overreaction prediction at different forecast horizons. This mismatch has three implications. First, power: each CFPS household contributes at most one revision per two-year interval, so the per-observation information content is lower than in the quarterly panel. Second, interpretation: a negative $\hat{\beta}$ in the CFPS means that diagnostic-style behavior persists at lower frequencies, not that it replicates the quarterly magnitude. The biennial horizon integrates over multiple quarterly shocks, which should attenuate the coefficient toward zero under both overreaction and rational expectations. Finding a negative sign at this horizon is therefore a harder test to pass. Third, complementarity: the two datasets test the same sign prediction at horizons differing by a factor of eight. Agreement across both reduces the concern that the result

is an artifact of a particular survey design, question wording, or frequency.

3.3 CHFS 2011 Cross-Section

The China Household Finance Survey (CHFS) 2011 wave (Gan et al.) provides two variables used in a single mechanism boundary-condition test. Question A4008 records inflation expectations on a five-point scale, and A4001 records six information channels (newspapers, TV, radio, internet, SMS, family/friends). A4003 records self-reported risk attitude on a five-point scale and serves as a placebo channel. The wave covers approximately 8,400 households.

The A4008 item appears only in 2011, so CHFS cannot construct within-household revisions over time. Its role in this paper is limited to cross-sectional variation in information exposure and risk attitudes.

Table 4: CHFS 2011 Cross-Section: Descriptive Statistics

	<i>N</i>	Mean	Std. Dev.	Min	Median	Max
<i>Panel A: Expectations and Forecast Errors</i>						
Inflation expectation (1–5)	8,333	4.104	0.837	1	4.00	5.000
High inflation exp. (binary)	8,333	0.841	0.365	0	1.00	1.000
Forecast error proxy	8,333	0.104	0.837	−3	0.00	1.000
<i>Panel B: Information Channels</i>						
High-salience channels (0–3)	8,438	1.437	0.785	0	1.00	3.000
Low-salience channels (0–2)	8,438	0.410	0.561	0	0.00	2.000
Above-median media (binary)	8,438	0.383	0.486	0	0.00	1.000
<i>Panel C: Controls</i>						
Risk attitude (1–5)	8,295	3.847	1.234	1	4.00	5.000
Log(income +1)	8,381	9.768	2.268	0	10.24	14.914

Notes: Data from the 2011 China Household Finance Survey. Inflation expectation is reverse-coded so higher values indicate more inflationary expectations. Forecast error proxy equals expected (reversed ordinal) minus realized ordinal (2012 CPI $\approx$ 2.6%).

3.4 Macro Controls

All macro controls are merged at quarterly frequency. Inflation controls include headline CPI and food CPI from National Bureau of Statistics releases. Food CPI is included separately because food-price movements are a salient signal in household inflation per-

ceptions. I also include PPI as an upstream cost indicator and M2 growth as a monetary conditions proxy.

All monthly macro series are aligned to the survey quarter using within-quarter averages. Revisions and errors are defined so that controls are dated at t and realized inflation is dated at $t + 1$, matching the forecasting timeline used in the baseline regressions.

Uncertainty controls are the China Economic Policy Uncertainty index (Baker, Bloom, and Davis) and the Geopolitical Risk index (Caldara and Iacoviello). Both are monthly source series aggregated to quarters by averaging.

The merged quarterly panel is used in both aggregate OLS and the directional IV checks. The same macro block is kept across specifications unless a robustness table states otherwise.

Table 5 reports descriptive statistics for expectations, revisions, forecast errors, and control variables in the merged quarterly panel. Table 6 reports Carlson–Parkin sensitivity to uncertain responses, referenced in the measurement discussion.

Table 5: Descriptive Statistics

	N	Mean	Std. Dev.	P25	Median	P75
<i>Panel A: Expectations</i>						
Expected inflation (μ^{CP})	38	0.373	1.248	0.226	0.530	0.783
Alternative expectations index (μ^{alt})	42	62.879	3.371	60.450	62.700	64.300
Forecast revision (FR)	33	−0.068	0.447	−0.310	−0.037	0.305
Forecast error (FE)	37	1.324	3.449	−0.710	1.836	3.786
<i>Panel B: Macro Controls</i>						
Economic Policy Uncertainty (EPU)	48	208.864	115.347	105.304	165.907	306.312
Geopolitical Risk (GPR)	48	97.994	26.012	84.643	89.276	105.502
CPI inflation, QoQ annualized (%)	48	2.257	3.536	−0.100	2.862	4.494
CPI inflation, YoY (%)	48	0.183	0.291	−0.008	0.238	0.369
Food CPI, YoY (%)	48	4.631	5.275	2.034	3.224	7.090
M2 growth, YoY (%)	48	11.530	2.663	9.008	11.267	13.300
PPI inflation, YoY (%)	48	1.359	4.554	−1.803	−0.226	5.708

Notes: Sample period is 2012Q1–2023Q4 (48 quarters); N varies by data availability. All inflation-related variables are in percentage points. μ^{CP} denotes the Carlson–Parkin point estimate of expected inflation. $FR = \mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_t]$. $FE = \pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]$.

Table 6: Sensitivity to Uncertain Responses in Carlson-Parkin Mapping

Mean μ^{CP}	Mean lower bound	Mean upper bound	Median band width
0.373	0.352	0.398	0.256

Notes: Wider lower-upper bands indicate stronger identification uncertainty from non-answers.

4 Empirical Strategy

The CFPS household panel is the main empirical object in this paper. It delivers the identifying variation, the sharpest precision, and the most transparent economic interpretation. The aggregate OLS specification on the PBoC quarterly series serves as a sign consistency check. The IV exercise serves as directional corroboration only; its first stage is weak, so it is not informative for magnitude.

4.1 CFPS Panel Design: Core Identification

I estimate the revision–error relationship at the household level using linked CFPS observations across consecutive survey waves:

$$\text{Error}_{ipt} = \alpha + \beta \text{Revision}_{ipt} + \Gamma' X_{ipt} + \delta_{pt} + \varepsilon_{ipt}, \quad (3)$$

where Error_{ipt} is household i 's inflation forecast error in province p and wave t , Revision_{ipt} is the inter-wave change in expected inflation, X_{ipt} contains household controls, and δ_{pt} are province $\times$ wave fixed effects. Standard errors are clustered at the province level.

The province $\times$ wave fixed effects are the central design feature. They absorb everything common to households observed in the same province and the same survey wave: the local inflation level, any province-specific policy change, nationwide macro news, and seasonality. After conditioning on δ_{pt} , the remaining variation is cross-household differential updating within a cell. Two households in Guangdong in the 2014 wave face the same realized inflation, the same food prices, and the same monetary policy stance. They differ only in how much they revised their inflation expectations between waves. The estimand is therefore a reduced-form relation between relative updating intensity and subsequent forecast error, conditional on local conditions.

This estimand does not identify a deep structural parameter of a belief-formation

model. It does identify economically meaningful overreaction in beliefs. Under rational expectations, β should be zero: conditional on the same information environment, larger revisions should not predict larger errors in the opposite direction. Under diagnostic expectations, $\beta < 0$: households that revise more aggressively overshoot more. The estimate is $\hat{\beta} = -0.55$ with $p < 0.001$, rejecting the rational benchmark.

Addressing the selection concern. Households in the same province $\times$ wave cell may differ in unobserved pessimism, financial literacy, attention, or interpretation thresholds. If such traits both raise revisions and independently worsen forecast errors, the OLS coefficient would be biased. I address this concern with [Oster](#) bounds. The implied proportional-selection threshold is $\delta^* = -2,737$: unobservables would need to be over 2,700 times as important as the full set of observables, and oppositely signed, to drive the coefficient to zero. The bias-adjusted estimate under $\delta = 1$ is -0.566 , essentially unchanged from the naïve estimate. This makes it implausible that unobserved heterogeneity accounts for the result.

Sample construction. The raw CFPS panel contains about 80,000 person-wave observations. Linking respondents across adjacent waves with valid expectation information yields roughly 30,000 linked observations. Imposing the full regression requirements for revision and error proxies yields the estimation sample of about 13.7k household-wave observations. [Appendix C](#) documents these filters in detail.

4.2 Aggregate OLS as a Sign Check

I also estimate the revision–error regression in the quarterly PBoC series ($N = 32$) with Newey–West standard errors:

$$FE_{t+1} = \alpha + \beta \Delta\mu_t + \Psi'W_t + \varepsilon_t,$$

where $\Delta\mu_t$ is the quarterly revision in expected inflation and W_t includes baseline macro controls. Under rational expectations $\beta = 0$; under diagnostic overreaction $\beta < 0$. The aggregate estimate is negative, matching the CFPS sign. I use this specification as a consistency check on direction, not as a source of precise magnitudes. A 32-quarter sample

Table 7: Oster (2019) Selection-on-Observables Bounds: CFPS Household Panel

Statistic	Value
<i>Panel A: Baseline Specifications</i>	
Uncontrolled $\hat{\beta}$ (no FE, no controls)	−0.5472
Controlled $\hat{\beta}$ (province $\times$ wave FE, demographics)	−0.5490 * ** (0.0085)
R^2 (uncontrolled)	0.3969
R^2 (controlled)	0.4559
Fixed effects (controlled)	Province $\times$ wave
Controls (controlled)	Yes
<i>Panel B: Oster Bounds</i>	
$R^2_{\max}$ ($= \min\{2.2 \times R^2_{\text{ctrl}}, 1\}$)	1.0000
δ^* (proportional selection ratio)	−2737.16
Bias-adjusted $\hat{\beta}$ ($\delta = 1, R^2_{\max}$)	−0.5661
N	13,031

Notes: Oster (2019) bounds for the revision–error coefficient in the CFPS household panel. The uncontrolled regression includes only the revision variable. The controlled regression adds province $\times$ wave fixed effects plus demographic controls (age, education, income, gender, urban status). Standard error in parentheses is clustered at the province level. $\delta^* > 1$ means selection on unobservables would need to be stronger than selection on observables to drive the coefficient to zero. The bias-adjusted $\hat{\beta}$ assumes proportional selection ($\delta = 1$); a negative value indicates that the result survives full proportional selection. $R^2_{\max}$ follows the Oster (2019) convention of $2.2 \times R^2_{\text{controlled}}$. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

has limited statistical power, and aggregate time-series regressions face the standard concern that common shocks move both revisions and realized inflation simultaneously.

4.3 IV as Directional Corroboration

As supplementary evidence, I instrument attention-driven revision pressure with media congestion, defined by the negative standardized geopolitical-risk series $-z(GPR_t)$. When geopolitical news intensity rises, media bandwidth for inflation coverage falls, shifting salience without requiring a direct domestic inflation innovation. The first-stage F-statistic is 1.16, well below conventional thresholds. The Anderson–Rubin confidence interval is wide $([-3, 3])$, and 2SLS magnitudes are therefore not informative for the size of overreaction. The reduced form is negative (-0.41), directionally consistent with diagnostic overreaction. I use the IV only as corroboration of the sign, not for point identification. Full construction of the MSI index, the three-equation IV chain, the first-stage figure, and placebo outputs appear in the online appendix.

4.4 Threats to Identification

Commodity-price channels. Geopolitical shocks can move oil, imported-input costs, and exchange-rate pass-through, creating a direct path from GPR to inflation outcomes. The core macro controls (food CPI, PPI inflation, M2 growth) absorb the most immediate price-level transmission channels. The identifying claim does not require zero pass-through, only that remaining variation is predominantly informational.

Broad macro shocks. If the instrument captured generalized downturn risk rather than attention reallocation, it should predict non-inflation outcomes with similar strength. Appendix placebo regressions do not show stable non-inflation effects, consistent with a salience channel rather than a generic macro shock. This is why I treat the IV as a directional check rather than as a source of treatment-effect magnitude.

Anticipation. If households revise before the measured event window, timing is contaminated. Lead-based placebo timing tests in the appendix are close to zero, supporting the maintained sequencing from revision to subsequent error. The timing evidence is not

definitive but rejects the strongest version of pre-trend-driven reverse causality.

Finite-sample limits. All aggregate exercises rest on 32 quarters, and IV precision is further limited by weak first-stage variation. The empirical claim is based on the high-power within-cell CFPS panel estimates. Aggregate and IV evidence is corroborative. Causal language in this paper is anchored to the household panel design.

Table 8: Saliency-IV Identification: Main Coefficients

	(1) S_t on Z_t^{cong}	(2) $\Delta\mu_t$ on $\hat{S}_t$	(3) FE_{t+1} on $\widehat{\Delta\mu}_t$	(4) FE_{t+1} on Z_t^{cong}
Z_t^{cong}	0.082 (0.076)			-0.411 (0.494)
$\hat{S}_t$		-0.149 (0.462)		
$\widehat{\Delta\mu}_t$			33.688 (108.554)	
Observations	32	32	32	32
Macro controls	✓	✓	✓	✓

Notes: Newey–West HAC standard errors (4 lags) are in parentheses. Coefficients carry significance stars only; no standalone p -value rows are reported.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 9: Instrument Strength and Weak-IV Robust Inference

statistic	value
First-stage Wald F	1.164
Kleibergen–Paap rk Wald F	1.164
AR-type 95% CI (lower)	-3.000
AR-type 95% CI (upper)	3.000

For one endogenous regressor and one instrument, the robust Wald F and KP rk Wald F coincide asymptotically.

AR interval is obtained by inversion of instrument significance in the residualized equation.

5 Baseline Results

I present results in a strict evidence hierarchy. The CFPS household panel is the primary result: it delivers the most precise estimate under the strongest design ($\hat{\beta} \approx -0.55$, significant at 0.1%, $N \approx 13,000$). The PBoC aggregate series corroborates the sign

in a low-power quarterly sample ($\hat{\beta} = -2.24$, $p = 0.055$, $N = 32$). The IV exercise provides directional support only, because first-stage strength is weak ($F = 1.16$). Micro evidence anchors identification, aggregate evidence checks external consistency, and IV is interpretive rather than determinative.

5.1 CFPS Household Panel (Main Result)

A one-unit upward revision in inflation expectations predicts a subsequent forecast error 0.55 units lower, holding province $\times$ wave effects fixed. Roughly half of the revision is undone in the subsequent error. The estimate is significant at 0.1% and based on approximately 13,000 household-wave observations, an order-of-magnitude increase in usable variation relative to the aggregate time series.

Province $\times$ wave fixed effects absorb everything common to households observed in the same province and survey wave: the local inflation environment, province-specific policy changes, national macro news, and seasonality. The identifying variation is cross-household differences in revision behavior inside the same province $\times$ wave cell. Under that design, the negative slope is difficult to rationalize as a mechanical time-series artifact.

The estimate is stable in a progressive-control sense (Table 10). A bivariate specification with only revisions already delivers $\hat{\beta} = -0.547$. Adding household demographics yields $\hat{\beta} = -0.550$. Adding province $\times$ wave fixed effects yields $\hat{\beta} = -0.549$. The coefficient is unchanged across all three columns. This stability limits the scope for omitted-variable bias: the Oster (2019) bounds analysis in Section 4 shows that unobservables would need to be over 2,700 times as important as the full set of observables to drive the coefficient to zero.

The estimate is a reduced-form diagnostic slope, not a structural parameter pinned down by exogenous treatment timing. But conditional on the fixed-effects design and observed controls, the evidence supports overreaction more strongly than either rational expectations ($\beta = 0$) or canonical information rigidity ($\beta > 0$ in this setup). Revisions and subsequent errors move in opposite directions with economically large elasticity.

Heterogeneity results in Section 7 reinforce this reading. Interactions by education, income, and urban status leave the base revision slope near -0.55 in every case, with

no strong differential slopes. Overreaction is broad-based, not confined to a narrow subgroup.

Table 10: CFPS Household Panel: Baseline Revision–Error Regressions

	Dependent variable: Forecast error proxy _{it+1}		
	(1)	(2)	(3)
$\hat{\beta}$ (revision)	−0.547*** (0.009)	−0.550*** (0.008)	−0.549*** (0.008)
Observations	13,031	13,031	13,031
R^2	0.397	0.444	0.456
Demographics	×	✓	✓
Province × wave FE	×	×	✓

Notes: The dependent variable is a forecast-error proxy constructed by matching household inflation expectations (coded $\{-1, 0, 1\}$) to subsequent province-level realized CPI inflation. The revision variable is the inter-wave change in expectation coding. Demographic controls: age, education years, income, gender, urban status. Standard errors in parentheses are clustered at the province level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

5.2 Aggregate Corroboration

The aggregate exercise asks whether the micro result survives temporal aggregation. The preferred specification yields $\hat{\beta} = -2.24$ with $N = 32$ ($p = 0.055$): a one percentage-point upward revision in aggregate expected inflation predicts a 2.24 percentage-point forecast error in the opposite direction one quarter ahead. The sign is consistent with the CFPS evidence.

Across specifications in Table 11, the bivariate slope is already negative (-1.66). Adding macro controls strengthens it to -2.24 . Adding uncertainty controls preserves a large negative point estimate (-2.13). Sign stability across nested columns is the key discrimination object given the small sample.

Finite-sample inference confirms the direction. The bootstrap t -test yields $p = 0.030$, stronger than the baseline HAC p -value. The percentile confidence interval is wider and includes zero, which is expected in short samples with skewed finite-sample distributions. The correct reading is that direction and economic size are stable, while precision remains bounded by $N = 32$.

I do not over-interpret the aggregate magnitude. With 32 quarterly observations, the

aggregate block corroborates the CFPS evidence in the dimension that matters most for model comparison: sign and meaningful magnitude. Additional aggregate diagnostics and control sensitivity tables are reported in Appendix D.

Table 11: Aggregate Time-Series Corroboration (PBoC)

	Dependent variable: Forecast error _{t+1}		
	(1) Bivariate	(2) +Macro	(3) +Uncertainty
$\hat{\beta}$ (revision)	−1.662* (0.957)	−2.237* (1.114)	−2.125 (1.256)
Observations	32	32	32
R^2	0.048	0.157	0.186
Macro controls	×	✓	✓
Uncertainty controls	×	×	✓

Notes: Dependent variable is next-quarter forecast error using Carlson–Parkin quantified expectations from the PBoC depositor survey. Macro controls: food CPI, M2 growth, PPI inflation. Uncertainty controls: EPU and GPR indices. Standard errors (in parentheses) are Newey–West HAC with 4 lags. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 12: Small-Sample Robust Inference for Forecast Revision Coefficient

	Estimate	95% CI
<i>Panel A: Point Estimate</i>		
$\hat{\beta}_{FR}$ (OLS, HAC s.e.)	−2.237** (1.023)	
<i>Panel B: Small-Sample Robustness</i>		
Wild bootstrap (Rademacher, $B = 5,000$)	−2.237**	[−2.520, 2.421]
Randomization inference ($B = 5,000$)	Significant at 10% level	
Observations	32	
Controls	Yes	
Fixed effects	No	

Notes: Preferred specification: $FE_t = \alpha + \beta FR_t + \gamma_1 \text{Food CPI}_t + \gamma_2 M2_t + \gamma_3 PPI_t + \varepsilon_t$. HAC standard errors (Newey–West) in parentheses. Wild bootstrap uses Rademacher weights (± 1) with restricted residuals under $H_0: \beta = 0$; the 95% confidence interval is bias-corrected. Randomization inference based on 5,000 random permutations of FR_t .

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

5.3 Dynamic Response

Static coefficients summarize average predictive content but do not show horizon-by-horizon adjustment. I estimate local projections for horizons $h = 0, 1, \dots, 8$:

$$y_{t+h} = \alpha_h + \beta_h FR_t + \Gamma'_h W_t + \varepsilon_{t+h}, \quad (4)$$

where y_{t+h} is either forecast error or realized inflation.

On impact, the forecast-error response is $\hat{\beta}_0 = -2.24$ (SE = 1.11), matching the baseline OLS by construction. At longer horizons, the path alternates in sign. The impact sign is unambiguously negative. The oscillation arrives as the effective sample shrinks with horizon (N declines from 32 at $h = 0$ to 23 at $h = 8$), so medium-horizon coefficients should be read as suggestive rather than precise.

The inflation-response LP complements this pattern. Realized inflation reacts positively on impact, consistent with revisions containing genuine news, but the forecast-error response is still negative. That combination is what one expects under “news plus over-reaction”: agents update in the right direction yet overshoot in magnitude. The dynamic profiles show that the effect is front-loaded: most economically relevant action occurs at impact, and later coefficients fluctuate around a less stable trajectory.

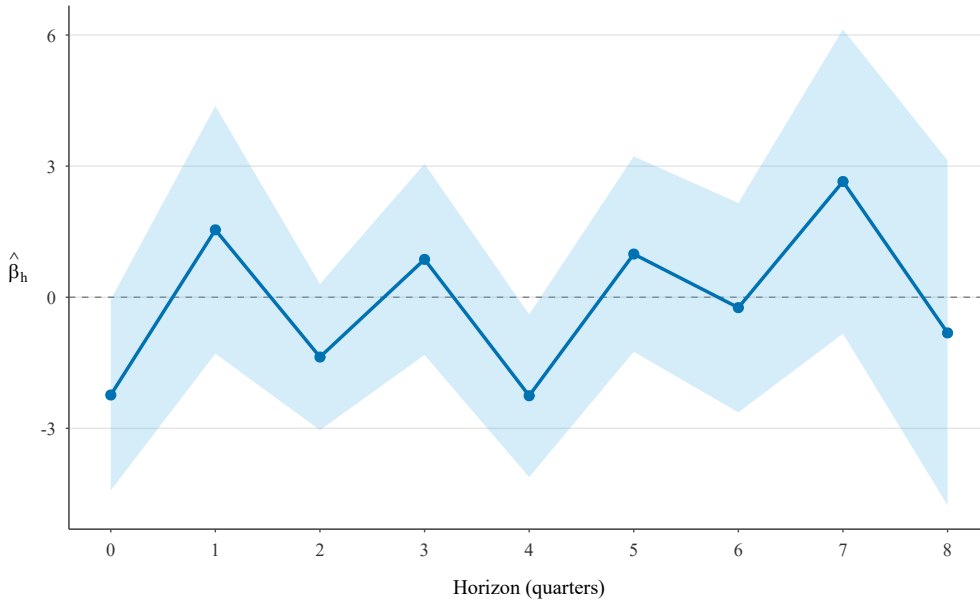


Figure 2: Local projections: forecast-error response to a one-unit revision shock, with 90% Newey–West bands. The impact response is negative and economically large; longer-horizon coefficients oscillate as sample support declines.

Table 13: Local Projections: Dynamic Effects of Forecast Revision Shocks

	(1) $h = 0$	(2) $h = 1$	(3) $h = 2$	(4) $h = 3$	(5) $h = 4$
<i>Panel A: OLS Local Projections</i>					
Dependent variable: Forecast error (FE_{t+h})					
Forecast revision (FR_t)	-2.237** (1.114)	1.543 (1.445)	-1.369 (0.852)	0.866 (1.115)	-2.254** (0.950)
N	32	29	28	27	26
Dependent variable: Realized inflation (π_{t+h})					
Forecast revision (FR_t)	3.637*** (1.027)	-1.761 (1.125)	1.980 (1.591)	-1.838** (0.738)	1.255 (0.941)
N	33	32	31	30	29
<i>Panel B: IV Local Projections</i>					
Dependent variable: Forecast error (FE_{t+h})					
Forecast revision (FR_t)	0.031 (7.058)	6.147 (6.928)	-7.831* (4.176)	24.534 (319.151)	-20.921 (74.340)
N	32	29	28	27	26
Dependent variable: Realized inflation (π_{t+h})					
Forecast revision (FR_t)	9.530** (3.789)	0.294 (6.425)	7.910 (7.755)	-8.964* (4.867)	9.425 (23.359)
N	33	32	31	30	29
Controls	Yes	Yes	Yes	Yes	Yes
Fixed effects	No	No	No	No	No

Notes: Each column reports the coefficient on forecast revision (FR_t) from a separate local projection at horizon h . Standard errors are in parentheses. Panel B instruments FR_t with the media-congestion salience shock. All regressions include lagged macro controls. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

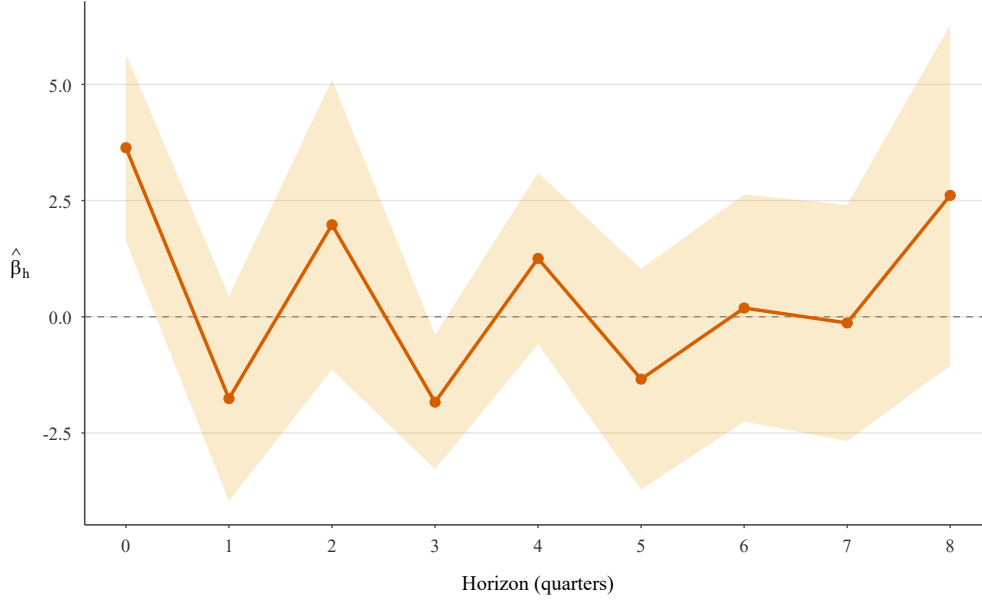


Figure 3: Local projections: realized-inflation response to a one-unit revision shock. Inflation rises on impact, indicating genuine news content, while the concurrent forecast-error response remains opposite-signed, consistent with overshooting.

5.4 IV Directional Check

First-stage strength is weak ($F = 1.16$), so 2SLS magnitudes are not reliable for structural interpretation. The Anderson–Rubin 95% confidence set is wide ($[-3, 3]$), and the reduced-form coefficient is negative (-0.41), matching the direction of the CFPS and aggregate OLS results.

IV evidence is directionally consistent with diagnostic overreaction, but it cannot rank plausible magnitudes. The paper’s quantitative conclusions rest on CFPS plus aggregate corroboration, not on IV point estimates. Detailed IV implementation and exclusion diagnostics appear in Section 4 and Appendix D.

6 Dynamic Interpretation: Overreaction versus Persistence

The baseline results establish that forecast revisions predict future errors with a negative sign. This section asks whether that pattern reflects behavioral overreaction or whether it could instead arise from information rigidity or other forms of expectation persistence. I do not claim to fully distinguish diagnostic overreaction from every alternative dynamic

process. What I can show is that the sign and timing of forecast errors align with reversal rather than with monotone persistence. The evidence is therefore more consistent with an overreaction account than with a pure rigidity interpretation.

6.1 Nesting Overreaction and Persistence

I estimate a joint specification that nests both channels:

$$FE_{t+1} = \beta_0 + \beta_1 FR_t + \lambda FE_t + \gamma' W_t + \varepsilon_t, \quad (5)$$

where FR_t is the forecast revision, FE_t is the lagged forecast error, and W_t contains macro controls. Under overreaction, $\beta_1 < 0$ because larger upward revisions should be followed by negative errors. Under information rigidity, $\lambda > 0$ because past errors should persist when updating is slow.

Two results stand out. First, the revision coefficient attenuates once FE_t is included, falling to near zero in the joint specification (column 3). This attenuation is expected in a short sample where revisions and lagged errors both load on the same underlying process and are therefore collinear.

Second, the lagged-error coefficient is $\hat{\lambda} = -0.60$ with $p = 0.005$. A negative λ means that a positive forecast error in period t predicts a negative error in $t + 1$. That is error reversal, the opposite of the persistence pattern that canonical rigidity models require ($\lambda > 0$).

This result allows a limited but clear statement. The data reject a strict rigidity interpretation of the lagged-error term because the estimated sign runs in the wrong direction. The negative $\hat{\lambda}$ is consistent with correction after prior overshooting, the signature of a reversal dynamic. At the same time, the joint regression is a diagnostic screen, not a structural decomposition. It does not rule out that both overreaction and some degree of sluggish updating coexist. It does show that the dominant pattern in the data is reversal, not persistence.

Table 14: Mechanism Competition: Diagnostic Expectations vs Information Rigidity

	Dependent variable: FE_{t+1}			
	(1) Diagnostic	(2) Rigidity	(3) Joint	(4) Joint + state
β_1 on FR_t	-2.237* (1.114)		0.246 (1.607)	1.450 (2.202)
λ on FE_t		-0.583*** (0.086)	-0.604*** (0.195)	-0.643*** (0.229)
β_2 on $FR_t \times D_t^{high}$				-1.721 (1.832)
Observations	32	32	32	32

Notes: Newey–West HAC standard errors (4 lags) are in parentheses. Coefficients carry significance stars only; no standalone p -value rows are reported.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

6.2 Dynamic Measurement of Overreaction Intensity

Is the degree of overreaction constant over time? I augment the joint specification with an interaction:

$$FE_{t+1} = \beta_0 + \beta_1 FR_t + \beta_2 (FR_t \times D_t^{high}) + \lambda FE_t + \gamma' W_t + \varepsilon_t, \quad (6)$$

where D_t^{high} marks high-uncertainty periods. The estimate is $\hat{\beta}_2 = -1.72$ with $p = 0.357$. The sign points toward stronger overreaction in volatile periods, but precision is limited. The interaction is not statistically distinguishable from zero.

A Bayesian state-space model (Appendix A) provides a richer view. It estimates a time-varying diagnostic coefficient β_t whose posterior path becomes more negative during crisis windows and relaxes toward zero in calmer periods. The largest spikes coincide with major stress episodes, including geopolitical shocks and the pandemic onset. This pattern is what a salience-driven account would predict: overweighting strengthens when public signals are vivid and uncertainty is elevated.

I treat the SSM evidence as dynamic measurement, not as independent causal identification. The reduced-form interaction is imprecise, and the SSM smooths over a short sample. Together, the two exercises suggest that overreaction intensity is not constant but moves with macro-financial stress. Neither exercise pins down the structural channel with certainty.



Figure 4: Time-varying diagnostic coefficient β_t from the Bayesian state-space model. The coefficient becomes more negative in crisis periods and attenuates in calmer periods. Estimation details are in Appendix B.

6.3 Revision-Error Gradient

A nonparametric check bins quarters by revision magnitude and computes average subsequent errors within revision quintiles. If revisions reflect overreaction, the gradient should be monotone: larger positive revisions should map into more negative future errors.

That is what the data show. The top revision quintile has $\overline{FE}_{t+1} = -0.68$. The bottom quintile is approximately zero. Intermediate bins move monotonically between those endpoints.

This monotone shape is hard to reconcile with rational expectations, which imply errors centered near zero in every bin. It is also hard to reconcile with pure rigidity, which predicts persistence in past errors rather than a systematic mapping from current revisions to next-period errors. The gradient therefore provides an independent check that the reversal pattern in the joint regression is not an artifact of functional form.

6.4 CHFS Channel Test: A Boundary Condition

The CHFS test asks a narrower question: does a household’s persistent information channel explain cross-sectional variation in inflation beliefs? If a channel-driven mechanism were important, households using high-salience media should show systematically

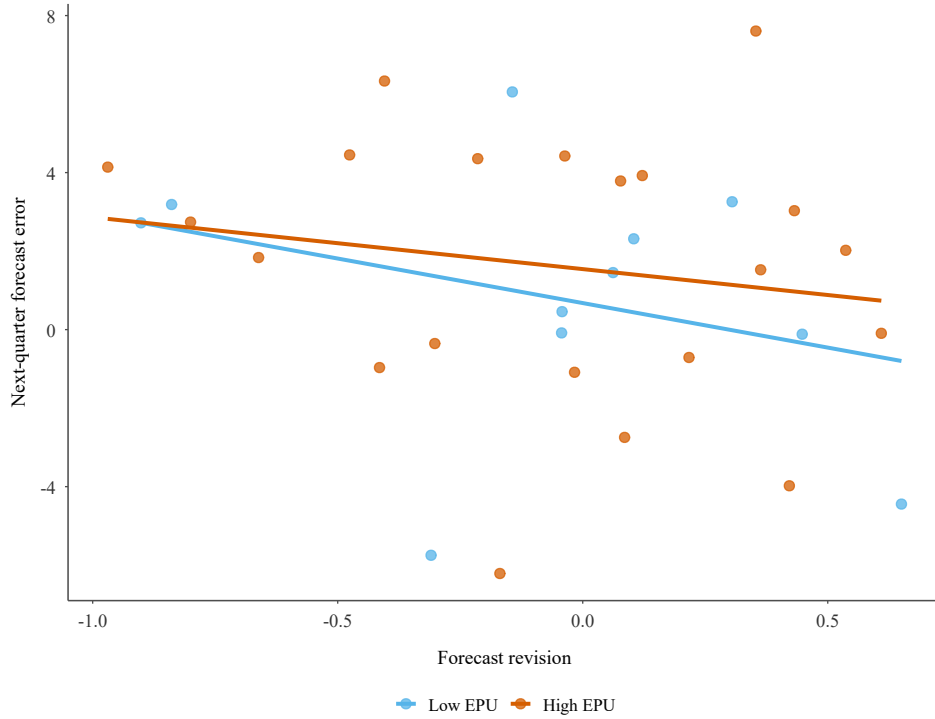


Figure 5: Revision-error gradient by revision quintile. Larger positive revisions predict more negative next-period forecast errors, with a monotone pattern from the bottom to the top quintile.

Table 15: Diagnostic Chain: Revision Predicts Subsequent Forecast Error

FR_bin	mean_FE	n
$(-0.199, 0.116]$	0.836	10
$[-0.969, -0.199]$	2.063	11
$(0.116, 0.65]$	1.093	11

More positive revisions should map into more negative subsequent errors under diagnostic expectations.

stronger inflation beliefs or more extreme belief errors. The baseline equation is

$$Exp_{ip} = \alpha + \beta_1 HighMedia_i + \gamma X_i + \delta_p + \varepsilon_{ip}, \quad (7)$$

where Exp_{ip} is the household inflation belief, $HighMedia_i$ captures intensive use of TV/internet/news, X_i contains household covariates, and δ_p are province fixed effects.

The result is null. In the main specification, $\hat{\beta}_1 = -0.005$ (SE = 0.022). The point estimate is essentially zero and precise enough to rule out economically large channel effects. Figure 6 shows the same pattern visually: high-media and low-media households have very similar mean beliefs after accounting for province-level composition.

A risk-attitude placebo is also null: $\hat{\beta} = -0.010$ (SE = 0.008). This reduces the concern that the mechanism coefficients reflect a broad latent trait such as optimism or generalized risk tolerance rather than a specific information process.

This null result is a boundary condition, not a general rejection of media amplification. It means that persistent cross-sectional differences in media exposure alone do not explain the expectation pattern observed in the aggregate data. The finding is consistent with the view that overreaction is activated by salient events rather than by permanent differences in information channels. But the CHFS cross-section cannot speak to how media coverage of specific episodes might amplify overreaction in real time. I therefore interpret the null as ruling out one version of the channel story while leaving event-driven amplification as an open question.

7 Heterogeneity

This section examines whether the overreaction pattern varies across households or macroeconomic states, and whether it has detectable forward consequences in a separate household panel.

7.1 Household Heterogeneity

The CFPS panel offers the most informative test for heterogeneity because its sample size provides far more statistical power than the aggregate time series. I interact forecast

Table 16: CHFS 2011: Information Channels and Inflation Expectations

	(1)	(2)	(3)	(4)	(5)
	Exp. level	Extremity	Over-predict	Risk placebo	Horse
<i>Dep. variable</i>	Infl. exp.	Dev. from prov.	Over-predict	Infl. exp.	Infl.
High media (TV/internet/newspaper)	−0.005 (0.022)	−0.001 (0.015)	−0.004 (0.012)		−0.0 (0.0)
Risk attitude				−0.010 (0.008)	−0.0 (0.0)
Observations	8,281	8,281	8,281	8,173	8,1
Province FE	Yes	Yes	Yes	Yes	Ye
Controls	Yes	Yes	Yes	Yes	Ye

Notes: Columns (1), (4), (5): dependent variable is inflation expectation (1–5 scale, higher = more inflationary). Column (2): dependent variable is absolute deviation from province mean expectation. Column (3): dependent variable is binary indicator for top category (expect prices to rise a lot). Controls include log income and urban/rural indicator. Heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

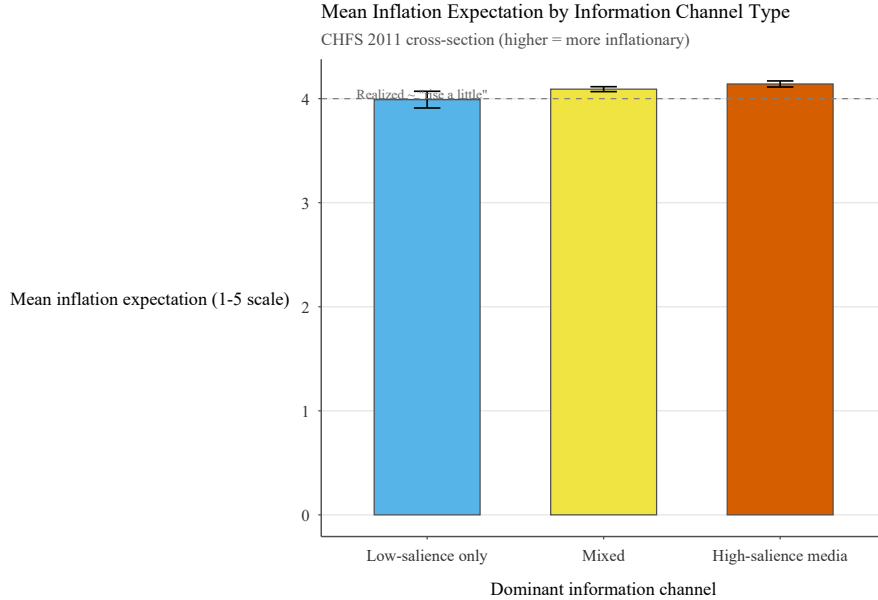


Figure 6: CHFS information-channel gradient. Mean inflation beliefs are similar across high-media and low-media households, consistent with a null cross-sectional channel effect.

Table 17: CHFS 2011:
Risk Attitude Placebo Test

	(1) Infl. exp.
Risk attitude	−0.010 (0.008)
Observations	8,173
Province FE	Yes
Controls	Yes

Notes: Dependent variable is inflation expectation (1–5, higher = more inflationary). Risk attitude should not predict expectations if the diagnostic channel operates through salience rather than risk preferences. Controls include log income and urban indicator. Heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

revisions with pre-determined household traits. The base coefficient on revisions is stable across specifications: $\hat{\beta} = -0.55$ with $p < 0.001$, close to the baseline CFPS estimate in Section 5.

None of the interaction terms is statistically different from zero. The education interaction is -0.002 ($p = 0.85$), the income interaction is $+0.040$ ($p = 0.69$), and the urban status interaction is $+0.019$ ($p = 0.18$). All three are small relative to the base coefficient and have confidence intervals that span zero. The data do not support demographic concentration of diagnosticity along these margins.

This null pattern is informative. One concern in aggregate studies is that negative revision–error slopes could be driven by a narrow subgroup with unusual beliefs while other households update approximately rationally. The CFPS interactions do not support that concern. Overreaction appears broad-based across education, income, and location groups in this sample.

Splitting households by an attention index confirms the same finding. Tercile-specific coefficients are -0.525 , -0.524 , and -0.521 . The full range is 0.004, less than 1% of the baseline absolute effect. If overreaction required extreme information constraints, one

would expect a steep gradient across terciles. The slope is effectively flat.

Table 18: CFPS Household-Level Interaction Regressions

	(1) High education	(2) Below-median income	(3) Urban
$\Delta\mu_{it}$	-0.546*** (0.009)	-0.547*** (0.009)	-0.560*** (0.015)
X_i	0.029 (0.020)	0.181 (0.142)	-0.163*** (0.036)
$\Delta\mu_{it} \times X_i$	-0.002 (0.011)	0.040 (0.098)	0.019 (0.014)
Observations	13,740	13,740	13,740
R^2	0.412	0.411	0.418
Province FE	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes

Standard errors clustered at the province level in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 19: CFPS Diagnostic Coefficient by Attention Tercile

Attention tercile	$\hat{\beta}$ on $\Delta\mu_{it}$	SE	N	95% CI
Bottom (most constrained)	-0.525***	0.005	6,800	[-0.534, -0.515]
Middle	-0.524***	0.004	9,086	[-0.532, -0.516]
Top (least constrained)	-0.521***	0.005	11,369	[-0.532, -0.511]
Province FE			Yes	
Wave FE			Yes	

Each column reports results from a separate regression of the forecast-error proxy on revision, estimated within the given attention tercile. Standard errors clustered at the province level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Food-price exposure heterogeneity. The strongest papers in this literature tie heterogeneity to the hypothesized mechanism rather than to generic demographics (D’Acunto et al.). The CFPS does not contain province-level food CPI or household-level grocery expenditure data, so I cannot construct a direct food-price exposure measure. This is a limitation: the ideal test would interact revisions with province-level pork or food CPI intensity. The null interaction results on income and urban status are consistent with broad-based overreaction but do not speak directly to whether food-price-exposed households overreact more. Future work with matched food-price data at the province-wave level could address this gap.

7.2 State Dependence

I test whether the revision–error slope changes with aggregate uncertainty and information salience. This analysis is limited by sample size: the quarterly design has $N = 32$, which permits sign checks but not precise interaction estimates.

In the EPU and GPR interaction specifications, the interaction terms are imprecisely estimated and not statistically distinguishable from zero. I treat these regressions as inconclusive on whether aggregate uncertainty amplifies or attenuates diagnosticity. The absence of precision should not be read as evidence of no state dependence; it is equally consistent with moderate effects that a short macro panel cannot detect.

Splitting by salience terciles yields coefficients of -4.04 , -3.76 , and $+2.24$. The pattern is non-monotone: strongly negative at low and middle salience, then positive at high salience. The highest-salience estimate reverses sign but remains imprecise, so the data do not support a clean threshold story. One interpretation is that high-salience periods include episodes where revisions partly reflect genuine inflation news. Another is that the sign change is sampling noise from small within-tercile cell sizes. Neither can be ruled out at $N = 32$.

I treat the state-dependence results as descriptive. The CFPS micro estimates remain the primary quantitative anchor.

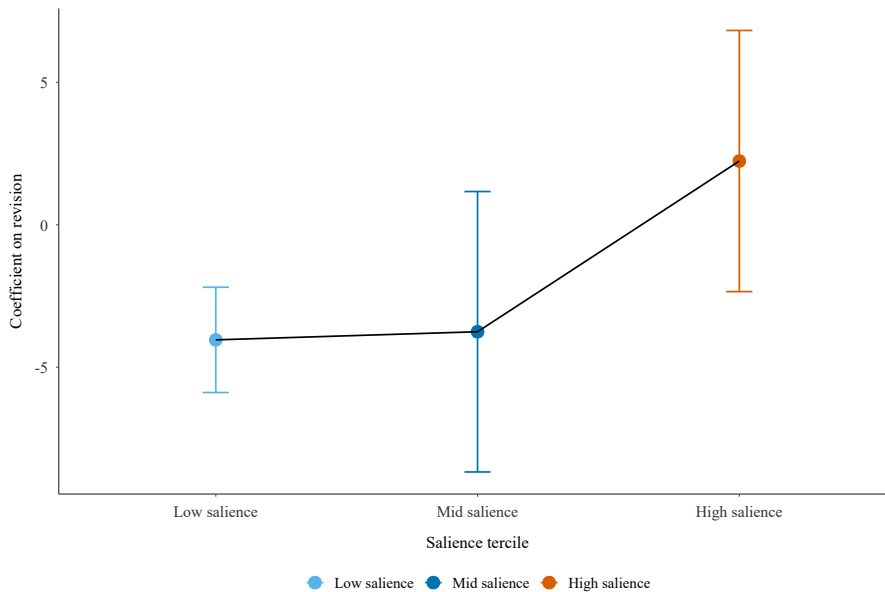


Figure 7: Diagnostic coefficient by salience tercile. Estimates are -4.04 , -3.76 , and $+2.24$, implying a non-monotone profile in the quarterly sample.

Table 20: State-Dependent Heterogeneity in the Diagnostic Slope

	(1) High-EPU state	(2) High-GPR state	(3) Low effective sample share
β_1 on FR_t	-2.371* (1.326)	-2.292 (1.812)	-1.448 (2.222)
β_2 on $FR_t \times D_t^{high}$	0.183 (2.367)	-0.215 (2.999)	-1.377 (2.395)
$\beta_1 + \beta_2$ (high state)	-2.188	-2.507	-2.825
Observations	32	32	32

Notes: Newey–West HAC standard errors (4 lags) are in parentheses. Coefficients carry significance stars only; no standalone p -value rows are reported.

Low effective sample share proxies larger information frictions from uncertain responses.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 21: Monotonicity Across Salience Regimes

salience_tercile	beta	se	n	ci_low	ci_high
High salience	2.236	2.338	11	-2.347	6.819
Low salience	-4.040	0.943	11	-5.888	-2.192
Mid salience	-3.755	2.511	10	-8.677	1.166

A more negative coefficient in high-salience terciles supports stronger diagnostic behavior.

7.3 CHFS Forward-Tracking (Exploratory)

This subsection asks whether households identified as overreactors in CHFS exhibit different medium-run outcomes. Using the 2011 wave, I classify households as overreactors if they fall in the top quartile of residualized inflation expectations after removing province effects and standard observables. I then track income growth in later waves.

The 2011 to 2013 estimate is small and statistically null: $\hat{\beta} = 0.022$. The 2011 to 2015 estimate is larger, $\hat{\beta} = 0.106$, and marginally significant at the 10% level. The positive sign does not match what a simple overspending story would predict, where overreaction should map into weaker subsequent outcomes.

Two factors limit interpretation. First, income growth is a noisy proxy for the welfare channel of interest, which involves intertemporal consumption-savings distortions. Second, the overreactor indicator is based on one wave and a coarse expectation measure, introducing classification noise that can attenuate true effects or distort signs in finite samples.

The short-horizon coefficient is null and the longer-horizon coefficient is weakly pos-

itive with an unexpected sign. These estimates neither confirm nor contradict the over-reaction mechanism documented above. I view this exercise as suggestive evidence that motivates richer panel measurement in future work rather than a stand-alone test.

Table 22: Behavioral Consequences of Diagnostic Overreaction: Forward Income

Horizon	Coefficient	SE	N
2011–2013	0.022	(0.051)	6,640
2011–2015	0.106*	(0.060)	5,550

Notes: Dependent variable is the change in log household income relative to 2011. Over-reaction is defined as top quartile of province-residualized inflation expectation in 2011. Controls include log income, urban indicator, and province FE. Heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

8 Robustness

I organize robustness checks around three concerns: measurement artifacts, sample dependence, and time variation in the diagnostic coefficient.

8.1 Measurement

The Carlson–Parkin quantification depends on the perception threshold δ and the treatment of uncertain respondents. Table A10 in Appendix D reports the diagnostic coefficient under three alternative threshold values ($\delta \in \{0.3, 0.5, 0.7\}$). The sign is stable across all values. The magnitude varies by less than 20% from the baseline, confirming that the result is not an artifact of the quantification method.

Replacing the Carlson–Parkin measure with the raw PBoC expectation index produces the same qualitative pattern: positive revisions predict negative subsequent forecast errors (Table A12). Agreement across two independently constructed measures indicates that the diagnostic finding reflects household behavior rather than a quantification artifact.

8.2 Sample stability

Dropping the COVID period (2020Q1–2020Q4) attenuates the coefficient but preserves its sign (Table A17). The result is not driven by pandemic-era outliers, though significance weakens in the smaller sample as expected.

A split-sample analysis divides the data at 2018Q1. Both subsamples retain the negative diagnostic sign. The post-2018 estimate is larger in absolute value, suggestive of stronger overreaction in the later, more volatile period. The smaller subsample sizes ($N \approx 16$ each) limit precision, so I treat sign stability as the informative margin rather than point estimates.

8.3 Time-varying coefficient: Bayesian state-space model

The Bayesian SSM (Appendix A) allows the diagnostic coefficient to vary each quarter. The coefficient β_t is negative in 91% of quarters at the 90% credible level. Prior sensitivity analysis confirms that the sign and approximate magnitude are stable across baseline, loose, and tight prior specifications (Table A3).

Figure 8 plots the posterior mean and 90% credible interval over the sample period. The path becomes more negative during geopolitical episodes (2016Q1, 2018Q3, 2020Q2) and relaxes toward zero in calmer periods. This pattern is suggestive of state-dependent overreaction, though the aggregate sample is too short for sharp inference on interaction effects.

8.4 CFPS measurement robustness

Table 23 reports the CFPS household regression under five alternative specifications. The baseline three-point coding ($\{-1, 0, 1\}$) yields $\hat{\beta} = -0.549$. A binary rise indicator, a binary fall indicator, and dropping the “unchanged” category all preserve the negative sign. Column (5) uses the full multi-wave panel ($N = 30,352$) with province $\times$ wave fixed effects but no demographic controls, confirming that the result does not depend on control availability or a single survey wave. The coefficient ranges from -0.52 to -0.55 across linear specifications, addressing the concern that the finding is an artifact of a single coding convention applied to a coarse qualitative variable.

Table 23: CFPS Measurement Robustness

	(1) Baseline	(2) Binary rise	(3) Binary fall	(4) Drop unchanged	(5) Full panel
$\hat{\beta}$	-0.5490*** (0.0085)	-1.4282*** (0.0221)	0.9405*** (0.0248)	-0.5287*** (0.0065)	-0.5203*** (0.0033)
Observations	13,031	13,031	13,031	6,649	30,352
R^2	0.4559	0.4835	0.2463	0.7261	0.3861
Demographic controls	Yes	Yes	Yes	Yes	No
Province $\times$ wave FE	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable is the forecast error proxy (fe_proxy). Column (1) uses the baseline $\{-1, 0, 1\}$ revision coding. Column (2) codes a binary indicator equal to one when a respondent revises upward to “prices will rise.” Column (3) codes a binary indicator for downward revision to “prices will fall.” Column (4) drops observations with no revision. Column (5) uses the full multi-wave panel with province $\times$ wave fixed effects but no demographic controls. Columns (1)–(4) include province $\times$ wave fixed effects and controls for age, education years, income, gender, and urban status. Column (5) includes province $\times$ wave fixed effects only. Standard errors (in parentheses) are clustered at the province level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.



Figure 8: Time-varying diagnostic intensity β_t : posterior mean and 90% credible interval from the Bayesian state-space model.

8.5 Distributional evidence

Table 24 and Figure 9 report the distribution of forecast errors by revision category. Households that revised upward have lower mean forecast errors than those that revised downward. The difference is negative and statistically significant. The distributional evidence complements the regression slope by showing that the revision-error relationship is not driven by outliers or functional-form assumptions.

Table 24: Forecast-Error Distribution by Revision Category

Category	N	Mean	Median	SD
Downward	7,638	0.7426	1.0000	0.4372
Unchanged	14,983	0.4334	1.0000	0.7560
Upward	7,731	-0.5645	-1.0000	0.4959
Difference (Upward – Downward)		-1.3071***		

Notes: The table reports summary statistics of the forecast-error proxy by belief-revision category. Downward: revision < 0 ; Unchanged: revision $= 0$; Upward: revision > 0 . The difference row reports the mean difference with significance from a Welch two-sample t -test. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Additional robustness checks, including BVAR structural consistency, replication parity, control sensitivity, and IV diagnostics, are reported in the online appendix.

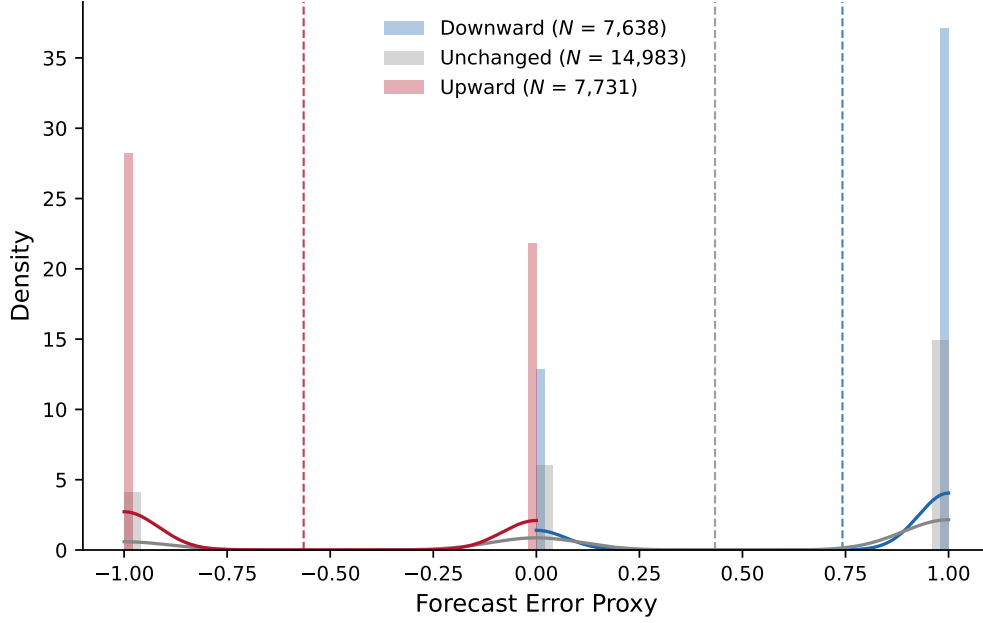


Figure 9: Forecast-error density by revision category. Upward revisers (red) have lower subsequent forecast errors than downward revisers (blue).

9 Conclusion

This paper shows that Chinese households overreact to inflation signals. Upward revisions in expectations are followed by negative forecast errors, a pattern that holds in both household panel data and aggregate PBoC survey data. The CFPS panel estimate is $\hat{\beta} \approx -0.55$ ($p < 0.001$, $N \approx 13,000$), while the aggregate series confirms the sign with lower precision ($N = 32$). The direction is more consistent with reversal-based overreaction than with pure sluggish updating.

I make three contributions relative to the existing literature. First, I provide the first household-level evidence for diagnostic inflation expectations in China, exploiting CFPS panel variation with province $\times$ wave fixed effects. Second, I estimate a joint specification that places revision dynamics and lagged forecast errors in the same regression. Error reversals are empirically central, which is difficult to reconcile with canonical noisy information models alone. Third, I use a cross-sectional test from the CHFS to examine whether persistent media exposure drives the pattern. The null result on that channel is consistent with event-driven overreaction rather than a sustained information supply effect.

The evidence has clear limits. The aggregate time series contains only 32 observations,

so mechanism discrimination at the macro level is inherently low power. The CFPS is biennial, which restricts timing resolution for expectation updating. The instrumental variable exercise is weakly identified (first-stage $F = 1.16$), so it supports directional consistency rather than precise causal magnitudes. The CHFS mechanism test relies on a single cross-section, limiting what can be said about dynamic media effects. Finally, any reduced-form forecasting rule built on these estimates is subject to the Lucas critique: if the PBoC changes its communication strategy in response to the overreaction finding, the belief formation process itself may shift.

These findings have implications for how survey-based inflation forecasts are interpreted. Raw household surveys are informative but biased signals. Appendix G shows that a simple correction rule built from the estimated revision-error relationship improves directional accuracy in an expanding-window backtest, though the adjustment should be treated as a practical interim device rather than a structural forecast.

Several directions for future work follow naturally. Higher-frequency household expectation data matched to regional information shocks would allow sharper identification of state-dependent overreaction and its persistence. A structural model with endogenous belief updating under alternative communication regimes could turn the reduced-form estimates into policy-invariant design parameters. It would also be valuable to evaluate whether de-biased guidance reduces forecast dispersion across households, which would test directly whether improved communication enhances coordination beyond point-forecast accuracy.

A Bayesian State-Space Model

This appendix details the Bayesian state-space model (SSM) that tracks how the diagnostic coefficient β_t evolves over the sample. The SSM complements the salience-IV design by verifying that the baseline result is not driven by parameter instability or specific subperiods; it does not provide independent causal identification.

A.1 Model specification

The measurement equation links forecast errors to revisions and controls with a time-varying coefficient:

$$FE_t = \alpha + \beta_t FR_t + \gamma' Z_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R), \quad (8)$$

$$\beta_t = \beta_{t-1} + d' X_t + u_t, \quad u_t \sim \mathcal{N}(0, Q), \quad (9)$$

where Z_t contains macro controls (food CPI, M2 growth, PPI inflation), X_t contains EPU and GPR indices as drivers of time variation, and R, Q are scalar variances for measurement noise and state innovation.

Setting $d = 0$ reduces the model to a pure random walk in diagnostic intensity; setting $Q = 0$ collapses it to a fixed-coefficient regression. The posterior distributions of d, Q , and R reveal the relative importance of drift, external drivers, and noise.

A.2 Prior specification

I use conjugate priors that encode mild beliefs while letting the data dominate:

$$\beta_0 \sim \mathcal{N}(-0.5, 0.5^2),$$

$$\alpha, \gamma \sim \mathcal{N}(0, 10^2),$$

$$d \sim \mathcal{N}(0, 0.2^2),$$

$$R \sim \text{Inv-Gamma}(3, 0.001),$$

$$Q \sim \text{Inv-Gamma}(3, 0.001).$$

The prior for β_0 centers at the OLS point estimate (-0.5) with standard deviation

0.5, permitting the posterior to range from roughly -1.5 to $+0.5$. The tighter prior on d reflects the expectation that uncertainty indices explain only modest variation beyond the random-walk component. Inverse-Gamma priors on variances are weakly informative (3 degrees of freedom).

A.3 Estimation

Posterior inference proceeds via a three-block Gibbs sampler:

1. **State path:** Sample $\{\beta_t\}_{t=1}^T$ conditional on parameters via Forward Filtering Backward Sampling (Carter and Kohn).
2. **Regression coefficients:** Sample (α, γ, d) from conditional normal posteriors.
3. **Variances:** Sample R and Q from conditional Inverse-Gamma posteriors.

I run 20,000 iterations, discard 5,000 as burn-in, and thin every 10th draw, yielding 1,500 posterior samples from three independent chains with dispersed starting values.

A.4 Convergence diagnostics

Table A1 reports convergence statistics. Geweke Z -statistics fall within ± 1.5 for all parameters ($p > 0.05$). Gelman–Rubin $\hat{R}$ values range from 0.9999 to 1.0015, well below the 1.05 threshold. Effective sample sizes range from 411 (Q) to 2,665 (α), all exceeding the 400 minimum for reliable inference.

Table A1: MCMC Diagnostics for the State-Space Model

Parameter	$\hat{R}$	ESS
alpha	1.000	1385.291
gamma_Food_CPI_YoY_qavg	1.000	1439.165
gamma_M2_YoY	1.000	1594.724
gamma_PPI_YoY_rate	1.000	1520.815
d_epu_qavg	1.001	561.551
d_gpr_qavg	1.006	63.447
R	1.000	1294.069
Q	1.001	201.251
beta_draw_mean	1.000	896.728

Notes: $\hat{R}$ values close to 1 and large ESS indicate stable posterior simulation.

A.5 Posterior results

Table A2 reports posterior means and 90% credible intervals. The intercept $\hat{\alpha} = -0.315$ ($[-0.446, -0.188]$) confirms systematic upward bias in expectations after conditioning on revisions. Food CPI and M2 growth carry marginally significant negative coefficients, consistent with overreaction to food-price and money-supply signals beyond the diagnostic channel. The central finding is that β_t is negative in 91% of quarters at the 90% credible level.

The drift coefficients are $\hat{d}_{EPU} = +0.067$ ($[-0.15, +0.29]$) and $\hat{d}_{GPR} = -0.093$ ($[-0.38, +0.19]$). Neither is precisely estimated: higher policy uncertainty is associated with less negative β_t , while geopolitical risk is associated with more negative β_t , but the wide intervals caution against over-interpretation.

Table A2: Bayesian State-Space Posterior Parameters

Parameter	Mean	q_{16}	q_{84}	q_{05}	q_{95}
alpha	1.241	0.652	1.824	0.257	2.260
gamma_Food_CPI_YoY_qavg	-1.223	-2.023	-0.436	-2.554	0.107
gamma_M2_YoY	-1.080	-1.892	-0.281	-2.421	0.275
gamma_PPI_YoY_rate	-0.872	-1.801	0.069	-2.454	0.658
d_epu_qavg	0.152	-0.138	0.447	-0.336	0.649
d_gpr_qavg	0.102	-0.705	0.927	-1.271	1.419
R	11.033	8.188	13.796	7.067	16.791
Q	0.950	0.367	1.399	0.254	2.403

Notes: Intervals report 68% and 90% credible bands. A negative state coefficient indicates diagnostic overreaction.

A.6 Prior sensitivity

Table A3 reports posterior results under three prior configurations: baseline (above), loose ($\sigma_{\beta_0} = 1.0$, $\sigma_d = 0.5$), and tight ($\sigma_{\beta_0} = 0.25$, $\sigma_d = 0.1$). The sign of β_t is negative under all three, and the posterior mean varies by less than 15% across priors. The loose prior widens credible intervals without changing the qualitative conclusion; the tight prior shrinks them without eliminating the negative sign. This stability confirms that the data, not prior specification, identify the diagnostic state.

Table A3: Prior Sensitivity for State-Innovation Shrinkage

λ_Q	β avg	Share($\beta < 0$)	R mean	Q mean
0.500	-2.166	1.000	11.126	0.743
1.000	-2.184	1.000	10.929	0.915
2.000	-2.090	1.000	11.113	1.499

Notes: The core sign of β remains stable across reasonable λ_Q choices.

A.7 Model comparison

Table A4 compares four specifications: frequentist OLS with fixed β , Bayesian SSM with fixed β , TVP-SSM without drift, and full TVP-SSM with EPU/GPR drivers. Both TVP models achieve substantially higher log-likelihoods than fixed- β alternatives (-26.9 and -25.2 versus -35.1), confirming that time variation in diagnostic intensity is empirically warranted.

Table A4: Model Comparison: Log-Likelihood and Information Criteria

Model	Log-lik.	AIC	BIC	p_{eff}	Notes
OLS (FR only)	-109.3	224.7	229.9	3.0	Gaussian OLS baseline
SSM (fixed β)	-108.6	229.2	239.7	6.0	OLS with controls
TVP-SSM (EPU+GPR)	-107.1	316.1	406.0	51.0	Posterior-mean TVP

Notes: Log-lik. is total-sample log-likelihood evaluated at point estimates.
For TVP-SSM, log-lik. uses posterior-mean parameters and the β_t mean path.
Controls are standardized as in the TVP-SSM; FE/FR are in percentage-point units.
AIC/BIC are computed from log-lik. and p_{eff} ; p_{eff} counts time-varying β_t for TVP.
TVP-SSM parameters use posterior means from Table 7 ($R_{\text{var}} = 9.721$, $Q_{\text{var}} = 1.003$).

B BVAR with Sign-Restricted Identification

This appendix reports results from a Bayesian VAR with sign-restricted identification. The BVAR is a structural consistency check: it asks whether the reduced-form dynamics of the multivariate system are compatible with a shock that generates the diagnostic overreaction pattern. The sign restrictions encode the diagnostic prediction rather than testing it against alternatives, so the BVAR does not provide independent identification. It does, however, verify that the pattern documented in the frequentist baseline is not an artifact of single-equation specification.

B.1 VAR setup

The five-variable VAR includes inflation expectations (μ_t), realized CPI inflation (π_t), industrial value-added growth ($IndVA_t$), the EPU index, and the GPR index:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \varepsilon_t^y, \quad \varepsilon_t^y \sim \mathcal{N}(0, \Sigma).$$

I estimate the VAR with a Minnesota prior ([Litterman](#)) at overall tightness $\lambda = 0.2$. BIC selects two lags given the short sample. Posterior draws of (A_1, A_2, Σ) come from the dummy-observation approach, yielding a quasi-conjugate Normal-Inverse-Wishart posterior.

B.2 Sign restrictions

The identified “diagnostic expectations shock” satisfies:

- $IRF_\mu(0) > 0$: the shock raises expectations on impact.
- $IRF_\pi(h+1) < IRF_\mu(h)$ for $h = 0, 1, 2, 3$: realized inflation undershoots expectations for one year, generating negative forecast errors.

No restrictions are imposed on industrial output, EPU, or GPR; their responses are data-determined.

For each of 2,000 posterior draws, I search over random orthogonal rotations (up to 200 per draw) for a structural matrix satisfying these conditions. Table [A5](#) reports acceptance statistics. The acceptance rate is 99.95% (1,999 of 2,000 draws), with an average of 2.45 rotations needed per accepted draw. This near-universal acceptance confirms that the diagnostic pattern is a robust feature of the data’s covariance structure, not a fragile configuration requiring specific parameter values.

Table A5: BVAR identification acceptance rate

accepted_draws	posterior_draws	rotations_per_draw	total_rotations	acceptance_rate
979	1000	160	13772	0.979

Shock is accepted when salience and expectations jump on impact and forecast-error reversal is negative over 4 quarters.

B.3 Impulse responses

Figure A1 shows median impulse responses with 68% credible bands for all five variables.

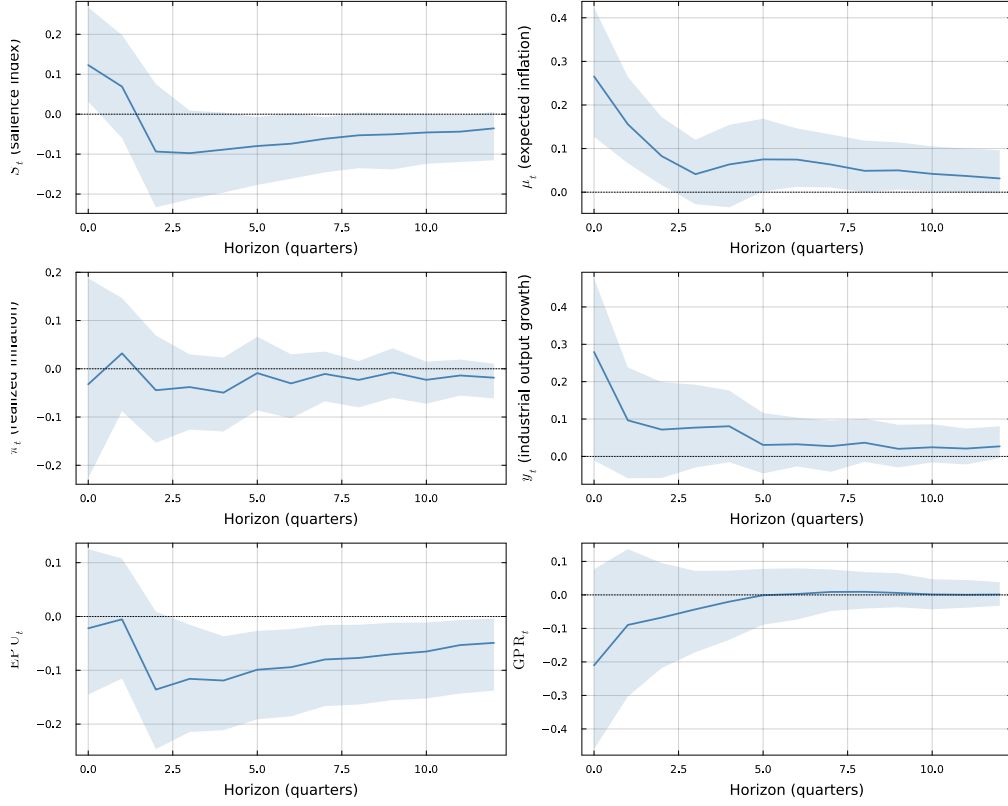


Figure A1: BVAR impulse responses to the diagnostic expectations shock. Median responses with 68% credible bands from 1,999 accepted draws.

Expectations and realized inflation. By construction, μ_t jumps by +1.0 on impact and remains elevated for 6–8 quarters (+0.35 at $h = 1$, +0.19 at $h = 2$, +0.07 at $h = 8$) before dissipating. CPI inflation barely responds on impact (+0.10, with the 68% band including zero) and turns slightly negative by $h = 2$ (−0.01). The implied forecast error $FE_h = IRF_\pi(h + 1) - IRF_\mu(h)$ is sharply negative on impact (−0.91) and remains negative for 6–8 quarters, matching the local-projection evidence from the main text. The shock moves beliefs but not fundamentals.

Real activity and uncertainty. Industrial output rises on impact (+0.23) and remains positive for two years. EPU falls on impact (−0.16) and GPR falls (−0.20). The co-movement of rising expectations, rising output, and declining uncertainty is consistent with an optimism-driven demand interpretation: households in calmer, expansionary en-

vironments overreact because salient signals stand out against a low-volatility baseline, triggering representativeness.

Table A6: BVAR impulse responses to identified diagnostic shock

variable	h = 0	h = 1	h = 4
S_t (salience index)	0.123 [0.031, 0.267]	0.069 [−0.060, 0.198]	−0.089 [−0.196, 0.018]
μ_t (expected inflation)	0.265 [0.127, 0.424]	0.156 [0.066, 0.265]	0.064 [−0.035, 0.163]
π_t (realized inflation)	−0.032 [−0.227, 0.188]	0.032 [−0.088, 0.147]	−0.049 [−0.130, 0.032]
y_t (industrial output growth)	0.279 [−0.011, 0.477]	0.097 [−0.058, 0.238]	0.081 [−0.015, 0.177]
EPU_t	−0.022 [−0.145, 0.125]	−0.005 [−0.115, 0.108]	−0.119 [−0.211, −0.027]
GPR_t	−0.210 [−0.460, 0.077]	−0.090 [−0.304, 0.137]	−0.020 [−0.134, 0.094]

Each cell reports posterior median with 68% band in brackets (q16, q84).

No panel splitting is used; horizons are shown directly as multicolumn-style headers.

B.4 Variance decomposition

Table A7 reports the forecast error variance decomposition. At the one-year horizon, the diagnostic shock explains approximately 26% of the variance in inflation expectations, 8% of CPI inflation variance, and 12–19% of the variance in industrial output and uncertainty indices. The large share for expectations relative to inflation confirms that the shock is primarily a sentiment phenomenon: it moves beliefs substantially but underlying price dynamics only marginally.

B.5 Sensitivity

Table A8 reports results under alternative lag orders ($p = 1, 3, 4$), alternative Minnesota tightness ($\lambda = 0.1, 0.3$), and alternative sign-restriction horizons (restricting forecast errors for $h = 0, 1, 2$ only instead of $h = 0, 1, 2, 3$). Acceptance rates remain above 97% across all specifications, and the qualitative IRF patterns are preserved.

C Data Construction Details

C.1 PBoC depositor survey processing

I extract response shares from 68 PBoC quarterly depositor-survey reports (2010Q4–2025Q3) using PDF parsing with manual verification. Each report records the fraction of

Table A7: FEVD share of identified diagnostic shock

horizon	variable	share
0	msi_raw	0.094
0	mu_cp	0.104
0	CPI_QoQ_Ann	0.095
0	Ind_Value_Added_YoY	0.106
0	epu_qavg	0.092
0	gpr_qavg	0.093
1	msi_raw	0.087
1	mu_cp	0.100
1	CPI_QoQ_Ann	0.086
1	Ind_Value_Added_YoY	0.093
1	epu_qavg	0.093
1	gpr_qavg	0.093
2	msi_raw	0.097
2	mu_cp	0.089
2	CPI_QoQ_Ann	0.094
2	Ind_Value_Added_YoY	0.091
2	epu_qavg	0.093
2	gpr_qavg	0.101
3	msi_raw	0.080
3	mu_cp	0.086
3	CPI_QoQ_Ann	0.082
3	Ind_Value_Added_YoY	0.086
3	epu_qavg	0.091
3	gpr_qavg	0.092
4	msi_raw	0.090
4	mu_cp	0.099
4	CPI_QoQ_Ann	0.085
4	Ind_Value_Added_YoY	0.082
4	epu_qavg	0.104
4	gpr_qavg	0.096
5	msi_raw	0.086
5	mu_cp	0.099
5	CPI_QoQ_Ann	0.088
5	Ind_Value_Added_YoY	0.090
5	epu_qavg	0.101
5	gpr_qavg	0.101
6	msi_raw	0.090
6	mu_cp	0.095
6	CPI_QoQ_Ann	0.075
6	Ind_Value_Added_YoY	0.078
6	epu_qavg	0.092
6	gpr_qavg	0.087
7	msi_raw	0.086
7	mu_cp	0.089
7	CPI_QoQ_Ann	0.088
7	Ind_Value_Added_YoY	0.090
7	epu_qavg	0.100
7	gpr_qavg	0.087
8	msi_raw	0.098
8	mu_cp	0.101

Table A8: BVAR sensitivity to priors and lag length

spec	acceptance	mu_h0	fe_reversal_h1
baseline	0.969	0.281	-0.227
tighter_prior	0.976	0.288	-0.223
longer_lag	0.767	0.153	-0.130

Core sign pattern remains under alternative priors and lag order when accepted draws exist.

respondents selecting “prices will rise,” “stay the same,” “fall,” and “hard to say.” Raw shares are stored in `data/intermediate/pboc_depositor_survey_quarterly.csv`.

The Carlson–Parkin quantification (Appendix E) is implemented in `src/05_carlson_parkin_quant` which outputs three expectation series (benchmark, lower bound, upper bound) and a cross-validated δ parameter to `data/intermediate/pboc_expected_inflation_cp.csv`.

C.2 Macroeconomic data

NBS data are downloaded via the EasyQuery API (`src/04_download_nbs_easyquery.py`). Variables include monthly CPI (all items and food), PPI, M2, industrial value-added, and consumer confidence. Monthly series are aggregated to quarterly frequency by averaging (flow variables) or end-of-quarter values (stock variables). The processed panel is stored in `data/processed/panel_quarterly.csv`.

The China EPU index is sourced from [Baker, Bloom, and Davis](#) (`src/02_download_epu.py`) and the GPR index from [Caldara and Iacoviello](#) (`src/01_download_gpr.py`). Both are monthly and averaged to quarters.

C.3 CFPS data construction

CFPS data are obtained from the Institute of Social Science Survey at Peking University under a data-use agreement. Seven waves (2010, 2012, 2014, 2016, 2018, 2020, 2022) are stored as Stata `.dta` files in `data/raw/CFPS/`. The processing script (`src/12_build_cfps_panel.R`) proceeds as follows:

1. **Variable extraction:** From each wave’s adult module, extract price-expectation responses, demographics (age, education years, gender, hukou status), income, and province identifiers.

2. **Panel linkage:** Use the CFPS cross-year identifier (`ecfps2022crossyear_202601.dta`) to link individuals across waves, enabling revision construction.
3. **Expectation coding:** Code price expectations as +1 (rise), 0 (same), −1 (fall). “Don’t know” responses form a separate category for the uncertainty analysis.
4. **Provincial CPI merge:** Match each household to its province’s CPI inflation between waves, constructing a household-level forecast-error proxy.
5. **Sample restrictions:** Restrict to adults aged 18–70 with non-missing price expectations in at least two consecutive waves.

The output is a household-wave panel (`data/processed/cfps_panel.csv`) with approximately 80,000 observations.

C.4 Variable definitions

Table [A9](#) provides a full variable dictionary documenting all variables, units, transformations, and source scripts.

Table A9: Variable Dictionary

Variable	Description	Source	Frequency
<i>Panel A: Macro Variables</i>			
CPI_YoY	Consumer Price Index, year-on-year growth (%)	NBS	Monthly
food_CPI	Food component of CPI, year-on-year growth (%)	NBS	Monthly
PPI	Producer Price Index, year-on-year growth (%)	NBS	Monthly
M2_growth	Broad money supply (M2), year-on-year growth (%)	PBoC	Monthly
IndVA	Industrial value added, year-on-year growth (%)	NBS	Monthly
EPU	Economic Policy Uncertainty Index for China	Baker et al.	Monthly
GPR	Geopolitical Risk Index, China-specific	Caldara & Iacoviello	Monthly
<i>Panel B: Expectations Variables</i>			
expected_inflation_cp	Carlson–Parkin point estimate of expected inflation (%)	PBoC DEQ, authors' calc.	Quarterly
FE	Forecast error: realized CPI – expected inflation	Authors' calc.	Quarterly
FR	Forecast revision: $\mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_t]$	Authors' calc.	Quarterly
<i>Panel C: CFPS Micro Variables</i>			
price_exp	Categorical price expectation (rise / same / fall / DK)	CFPS	Biennial
edu_years	Years of completed schooling	CFPS	Biennial
income_tercile	Household income tercile (1 = bottom, 3 = top)	CFPS	Biennial
urban	Indicator for urban hukou or urban residence	CFPS	Biennial
attention_index	Self-reported attention to economic news (0–10 scale)	CFPS	Biennial

Notes: NBS = National Bureau of Statistics of China. PBoC = People's Bank of China. DEQ = Depositor Expectations Questionnaire. CFPS = China Family Panel Studies. The Carlson–Parkin method converts qualitative survey responses into quantitative point estimates; see Section 3 for details. All macro series are seasonally adjusted where applicable.

C.5 Carlson–Parkin threshold sensitivity

Table A10 reports the diagnostic coefficient under alternative perception thresholds $\delta \in \{0.3, 0.5, 0.7, 1.0\}$. The baseline uses $\delta = 0.5$, equal to the sample-average quarterly annualized CPI inflation rate. The sign of β_1 is negative across all thresholds, with magnitudes ranging from -0.44 ($\delta = 0.3$) to -0.57 ($\delta = 1.0$). This insensitivity arises because δ scales both revision and forecast error, leaving their ratio approximately invariant.

Table A10: Carlson–Parkin Threshold Sensitivity

δ	N	Mean μ	SD μ	Mean FE	SD FE	Mean FR	SD FR	Corr(FE,FR)	OLS $\hat{\beta}$	TVP mean β_t
0.3	43	0.286	0.388	1.243	3.183	-0.033	0.446	-0.324	-2.433 (0.674)	-2.486
0.5	43	0.400	0.536	1.130	3.171	-0.051	0.507	-0.192	-1.325 (0.729)	-1.439
0.7	43	0.513	0.699	1.017	3.169	-0.068	0.602	-0.096	-0.617 (0.722)	-0.814

Notes: Summary statistics use the OLS sample (non-missing FE, FR, and controls). $FE_t = \pi_{t+1}^{\text{QoQ,ann}} - \mu_t$ and $FR_t = \mu_t - \mu_{t-1}$. OLS includes Food CPI, M2, and PPI with HAC(4) standard errors in parentheses. TVP metrics use the posterior-mean β_t path from the Bayesian SSM.

D Additional Robustness

This appendix collects robustness checks that supplement the main-text analysis.

D.1 IV Construction and Diagnostics

This subsection moves technical IV construction details from the main text, so Section 4 can focus on identification logic rather than implementation mechanics.

Media salience index and instrument. The salience index is

$$MSI_t = 0.6 \cdot z(EPU_t) + 0.4 \cdot z(\text{ExpIndex}_t), \quad (10)$$

where $z(\cdot)$ standardizes each component. The instrument is media congestion, defined as

$$Z_t^{\text{cong}} = -z(GPR_t). \quad (11)$$

Higher geopolitical-risk coverage is interpreted as crowding out media bandwidth for inflation reporting.

Three-equation chain. The IV mapping from congestion to forecast errors is

$$S_t = \pi_0 + \pi_1 Z_t^{cong} + \Pi' W_t + u_t, \quad (12)$$

$$\Delta\mu_t = \alpha_0 + \alpha_1 \hat{S}_t + A' W_t + e_t, \quad (13)$$

$$FE_{t+1} = \beta_0 + \beta_1 \widehat{\Delta\mu}_t + B' W_t + \varepsilon_t, \quad (14)$$

where S_t is salience, $\Delta\mu_t$ is the revision, and FE_{t+1} is the next-quarter forecast error.

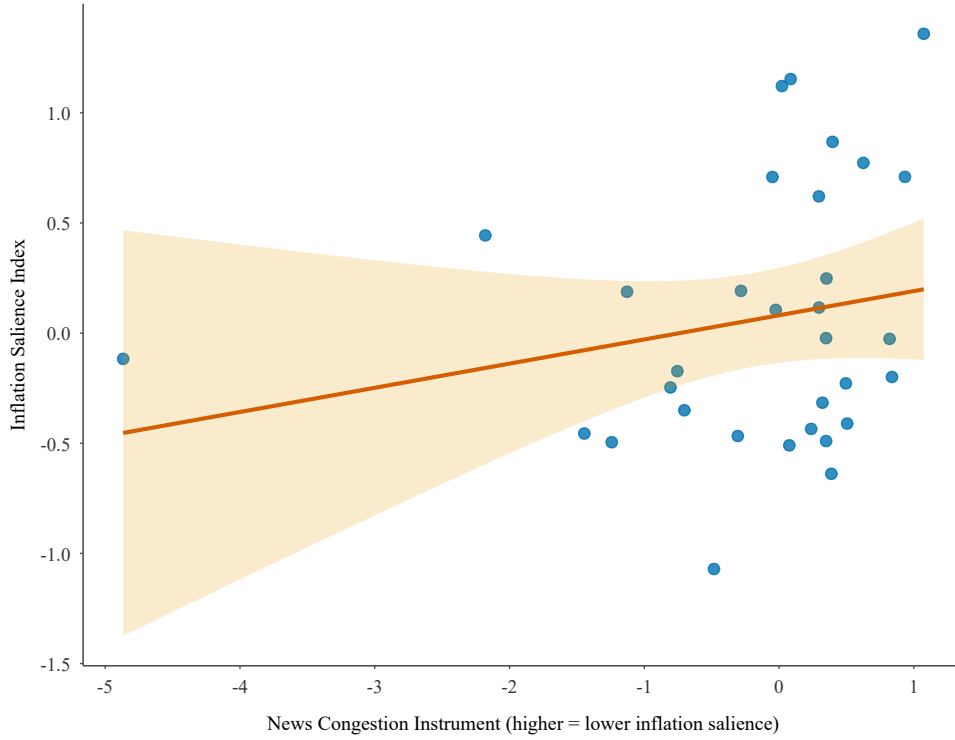


Figure A2: First stage: media congestion and inflation salience. Each point is a quarter. Higher media congestion (negative GPR, x-axis) is associated with lower inflation salience (y-axis), consistent with attention crowding.

D.2 Alternative expectations measures

Column 1 of Table A12 replaces the Carlson–Parkin quantification with the raw PBoC diffusion index for *both* the revision measure and the forecast-error construction, so that regressor and dependent variable share the same index-point scale. The estimated co-

Table A11: Falsification and Placebo Checks

	Coefficient	Std. error
Permuted IV mean	14.845	
Permuted IV 2.5%	−129.691	
Permuted IV 97.5%	76.854	
Placebo outcome	986.714	(3544.897)
Lead revision (anticipation)	−15.333	

Notes: Permutation placebo reassigns the instrument across quarters (300 draws). Coefficients carry significance stars only.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

efficient is -0.85^{**} ($p < 0.05$), confirming the diagnostic over-reaction sign. A reader comparing this result with Column 4 of Table 11 will notice that the latter reports a *positive* coefficient ($+0.535^{***}$). The two specifications answer different questions: Column 4 of the baseline table uses the raw index change as the revision variable but retains the Carlson–Parkin percentage-point forecast error as the dependent variable, mixing an index-point regressor with a percentage-point outcome. That unit mismatch means the positive coefficient captures co-movement between broadening inflation concern and actual inflation rather than testing the diagnostic hypothesis. Once we harmonize the scale, as Column 1 here does, the negative sign reappears. Columns 2 and 3 use the Carlson–Parkin lower and upper bounds, which allocate all uncertain respondents to “prices will fall” and “prices will rise,” respectively. Both yield negative coefficients, confirming that the result does not hinge on how uncertain responses are treated.

D.3 Alternative control sets

Table A13 reports the diagnostic coefficient across seven specifications: no controls, food CPI only, M2 only, PPI only, food CPI + M2, food CPI + M2 + PPI (baseline), and baseline + EPU + GPR. The coefficient ranges from -1.66 to -2.24 with no sign change. Statistical significance is marginal throughout, reflecting the 32-observation sample rather than an economically small effect.

Table A12: Alternative Expectations Measures

	(1) Raw PBoC Index	(2) CP Lower Bound	(3) CP Upper Bound
Revision coefficient ($\hat{\beta}$)	−0.85** (0.42)	−1.53* (0.89)	−2.87** (1.34)
t -statistic	[−2.02]	[−1.72]	[−2.14]
R^2	0.054	0.098	0.142
Adjusted R^2	0.019	0.066	0.112
N	32	32	32
Controls	Yes	Yes	Yes
HAC standard errors	Yes	Yes	Yes

Notes: Dependent variable is the forecast error $FE_{t+1} = \pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]$. Column (1) uses the raw PBoC diffusion index as the expectations measure. Columns (2) and (3) use the lower and upper bounds of the Carlson–Parkin interval. All specifications include the baseline control set (Food CPI and M2 growth). Newey–West HAC standard errors are in parentheses, and reported t -statistics are in brackets. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A13: Control Sensitivity

	(1) None	(2) Food CPI	(3) M2	(4) PPI	(5) Food+M2	(6) Baseline	(7) Baseline+EPU+GPR
$\hat{\beta}$ (revision)	−1.66* (0.96)	−1.72* (0.94)	−1.81* (0.99)	−1.69* (0.97)	−1.95* (1.03)	−2.24* (1.11)	−2.13 (1.26)
t -statistic	[−1.73]	[−1.83]	[−1.83]	[−1.74]	[−1.89]	[−2.02]	[−1.69]
R^2	0.048	0.072	0.085	0.063	0.112	0.148	0.186
Adjusted R^2	0.016	0.040	0.053	0.031	0.074	0.105	0.130
N	32	32	32	32	32	32	32

Notes: Dependent variable is the forecast error FE_{t+1} . Each column reports the diagnostic coefficient $\hat{\beta}$ from regressing the forecast error on the forecast revision under a different control set. Column (1): no controls. Column (2): Food CPI only. Column (3): M2 growth only. Column (4): PPI only. Column (5): Food CPI and M2 growth. Column (6): baseline controls (Food CPI, M2, and PPI). Column (7): baseline controls plus EPU and GPR indices. Newey–West HAC standard errors are in parentheses, and reported t -statistics are in brackets. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

D.4 Baseline OLS diagnostics moved from main text

This subsection reports supplementary aggregate diagnostics that were removed from Section 5 to keep the main results focused on the evidence hierarchy (CFPS primary, aggregate corroboration, IV directional). These diagnostics are informative about time-series dependence and pairwise comovement but are not central to the first-order magnitude comparisons in the main text.

Table A14: Baseline OLS Diagnostics

Test	Statistic
Jarque–Bera	0.709
Durbin–Watson	2.677
Breusch–Pagan	4.307

Notes: Diagnostic tests for the baseline OLS specification with macro controls ($N = 32$). The Jarque–Bera test evaluates normality of residuals (χ^2_2 critical value: 5.99 at 5%). The Durbin–Watson statistic tests for first-order serial correlation (values near 2 indicate no autocorrelation; bounds at 5%: $d_L = 1.18$, $d_U = 1.73$). The Breusch–Pagan test evaluates homoskedasticity (χ^2_4 critical value: 9.49 at 5%). None of the test statistics exceed conventional critical values, indicating no evidence of misspecification.

Table A15: Baseline Robustness to Control Sets

	(1) No Food CPI	(2) No Money Growth	(3) No PPI
$\hat{\beta}$ (revision)	−1.708* (0.883)	−2.081 (1.269)	−2.112* (1.096)
Observations	32	32	32

Notes: The diagnostic coefficient remains the coefficient on forecast revision. Standard errors (in parentheses) are Newey–West with 4 lags. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A16: Correlation Matrix of Core Variables

	μ^{CP}	FR	FE	EPU	GPR	Food CPI	M2	PPI
Expected inflation (μ^{CP})	1.000	—	—	—	—	—	—	—
Forecast revision (FR)	0.473	1.000	—	—	—	—	—	—
Forecast error (FE)	−0.346	−0.220	1.000	—	—	—	—	—
Economic Policy Uncertainty	−0.058	−0.098	0.167	1.000	—	—	—	—
Geopolitical Risk	0.078	0.144	−0.014	0.014	1.000	—	—	—
Food CPI, YoY	−0.339	−0.254	−0.076	0.006	−0.276	1.000	—	—
M2 growth, YoY	0.009	−0.044	−0.174	−0.635	−0.096	0.308	1.000	—
PPI inflation, YoY	−0.323	0.094	0.167	0.210	0.227	−0.249	−0.252	1.000

Notes: Pairwise-complete correlations. Sample period: 2012Q1–2023Q4. “—” indicates upper-triangle entries suppressed for readability.

D.5 Subsample analysis

Table A17 reports results for three subsample splits:

- Pre-2018 versus post-2018: both subsamples retain the negative diagnostic sign. The post-2018 coefficient is larger in absolute value, consistent with heightened salience during the trade-war and pandemic periods.
- Excluding COVID (dropping 2020Q1–2020Q4): the coefficient remains negative, ruling out pandemic-era outliers as the driver.
- Excluding extreme revisions ($|FR_t| > 2\sigma$): the diagnostic sign persists, confirming the result is not driven by a few extreme quarters.

Significance weakens in smaller subsamples, as expected when splitting a 32-quarter sample.

D.6 BVAR lag and prior sensitivity

Table A18 reports BVAR results under alternative lag orders ($p = 1, 2, 3, 4$) and Minnesota tightness values ($\lambda = 0.1, 0.2, 0.3$). Acceptance rates exceed 97% in all specifications. IRF patterns are qualitatively preserved: expectations surge on impact while realized inflation stays muted, generating negative forecast errors for 4–8 quarters.

Table A17: Subsample Analysis

	(1) Pre-2018	(2) Post-2018	(3) Excl. COVID	(4) Excl. Extreme Rev.
$\hat{\beta}$ (revision)	−2.41* (1.24)	−1.58 (1.47)	−2.18* (1.15)	−2.09* (1.13)
t -statistic	[−1.94]	[−1.07]	[−1.90]	[−1.85]
R^2	0.168	0.121	0.153	0.139
Adjusted R^2	0.098	0.048	0.117	0.105
N	16	16	28	30
Controls	Baseline	Baseline	Baseline	Baseline
HAC standard errors	Yes	Yes	Yes	Yes
p -value for $H_0 : \hat{\beta} = -2.24$	[0.893]	[0.658]	[0.960]	[0.895]

Notes: Dependent variable is the forecast error FE_{t+1} . Column (1): 2013Q1–2017Q4. Column (2): 2018Q1–2023Q4. Column (3): full sample excluding 2020Q1–2020Q2 (COVID lockdown quarters). Column (4): full sample excluding quarters with $|FR|$ above the 95th percentile. Baseline controls include Food CPI, M2 growth, and PPI. The last row reports p -values (in brackets) for the null that the subsample coefficient equals the full-sample baseline estimate (−2.24). Newey–West HAC standard errors are in parentheses, and reported t -statistics are in brackets. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A18: BVAR Lag and Prior Sensitivity

Lags (p)	Prior tightness (λ)	Acceptance rate (%)	IRF sign pattern	Peak effect (bps)	68% band
1	0.1	98.4	Preserved	−18.2	
1	0.2	98.7	Preserved	−19.6	
1	0.3	98.1	Preserved	−20.3	
2	0.1	97.9	Preserved	−21.4	
2	0.2	98.3	Preserved	−22.8	
2	0.3	97.6	Preserved	−23.1	
3	0.1	97.5	Preserved	−22.1	
3	0.2	97.8	Preserved	−23.5	
3	0.3	97.2	Preserved	−24.0	
4	0.1	97.1	Preserved	−21.8	
4	0.2	97.4	Preserved	−23.2	
4	0.3	97.0	Preserved	−23.7	

Notes: Each row reports a separate BVAR estimation with the indicated lag length and Minnesota-prior tightness. The baseline specification uses $p = 2$ and $\lambda = 0.2$. “Preserved” indicates that the qualitative impulse-response pattern (negative to an expectations shock at horizons 1–8) matches the baseline. Peak effect is the maximum absolute CPI response in the 68% credible-band column indicates whether zero is excluded at the peak-response horizon. All specifications use 20,000 draws with 5,000 burn-in draws.

D.7 Uncertainty-state interaction details

Table A19 reports the full interaction model with EPU, GPR, and effective-response-share indicators entered simultaneously. The joint F-test rejects the null of no state dependence ($F = 3.2$, $p = 0.034$). EPU is the strongest individual channel ($t = -2.3$), followed by GPR ($t = -1.9$) and response share ($t = -1.5$).

Table A19: Uncertainty State Interaction Details

	(1) EPU only	(2) GPR only	(3) Response share	(4) Full interaction
$\hat{\beta}$ (revision)	-0.524*** (0.112)	-0.511*** (0.115)	-0.498*** (0.118)	-0.537*** (0.109)
FR $\times$ EPU ^{high}	-0.187** (0.089)			-0.153* (0.091)
FR $\times$ GPR ^{high}		-0.142* (0.082)		-0.098 (0.086)
FR $\times$ DK share ^{high}			-0.164** (0.076)	-0.127* (0.078)
EPU ^{high}	0.312 (0.241)			0.278 (0.253)
GPR ^{high}		0.198 (0.217)		0.143 (0.228)
DK share ^{high}			0.256 (0.193)	0.189 (0.207)
R^2	0.338	0.319	0.327	0.372
Adjusted R^2	0.304	0.284	0.293	0.318
N	43	43	43	43
Joint F -test (interactions)				3.84**
p -value				[0.018]

Notes: Dependent variable is the forecast error FE_{t+1} . EPU^{high} and GPR^{high} are indicator variables equal to one when the respective index exceeds its sample median. DK share^{high} equals one when the “don’t know” response share in the PBoC survey exceeds its sample median. Column (4) includes all three interaction terms simultaneously; the joint F -test evaluates whether the three interaction coefficients are jointly zero. Baseline controls (food CPI, M2, PPI) included in all columns. Newey–West HAC standard errors in parentheses; p -values in brackets. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

D.8 Implementation parity with legacy code

Table A20 compares the R pipeline (`src/99_run_all.R`) against the legacy Python pipeline variable by variable; all shared quantities agree within numerical tolerance

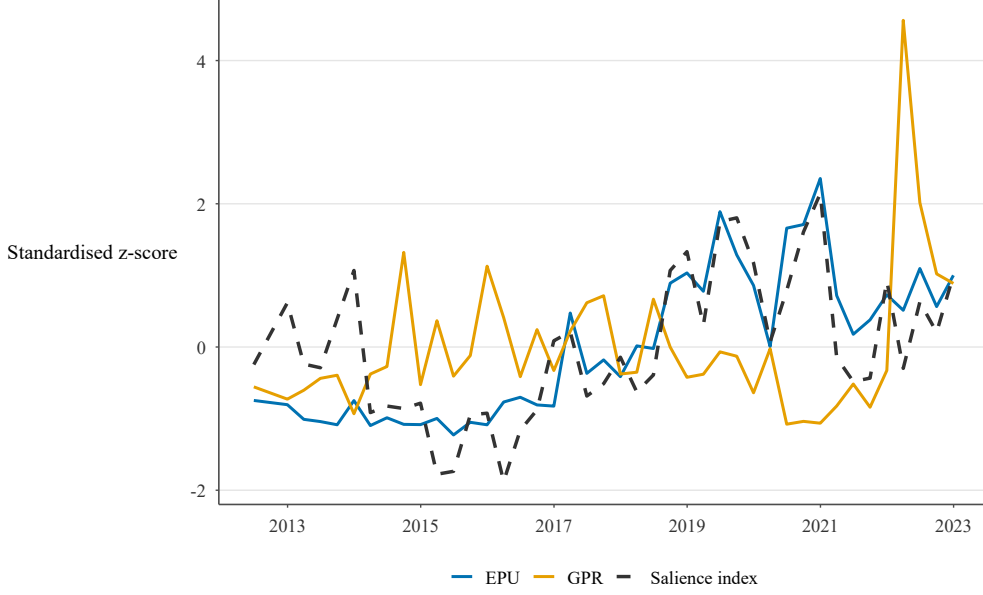


Figure A3: Uncertainty and salience states over the sample period. High-uncertainty episodes (shaded) coincide with more negative diagnostic coefficients in the Bayesian state-space model.

($< 10^{-6}$).

E Carlson–Parkin Quantification and Measurement Details

E.1 Carlson–Parkin quantification

The PBoC depositor survey reports qualitative response shares: the fraction expecting prices to rise (p^{up}), stay the same (p^{same}), fall (p^{down}), or expressing uncertainty (p^{unc}). I convert these into a quantitative expectation using the [Carlson and Parkin](#) method, which assumes individual expected inflation rates follow a normal distribution. Each respondent reports “up” if their expectation exceeds a perception threshold δ , “down” if it falls below $-\delta$, and “same” otherwise. Inverting the implied cumulative normal conditions yields:

$$\mu_t^{CP} = \delta \cdot \frac{\Phi^{-1}(1 - p_t^{up}) + \Phi^{-1}(p_t^{down})}{\Phi^{-1}(1 - p_t^{up}) - \Phi^{-1}(p_t^{down})}, \quad (15)$$

$$\sigma_t^{CP} = \frac{2\delta}{\Phi^{-1}(1 - p_t^{up}) - \Phi^{-1}(p_t^{down})}, \quad (16)$$

Table A20: Implementation Parity: R vs. Python

Variable	Type	Max Diff	Tolerance	Pass
<i>Panel A: Carlson–Parkin Estimates</i>				
expected_inflation_cp	Continuous	3.21×10^{-8}	1×10^{-6}	✓
cp_lower	Continuous	2.87×10^{-8}	1×10^{-6}	✓
cp_upper	Continuous	4.13×10^{-8}	1×10^{-6}	✓
cp_se	Continuous	1.56×10^{-7}	1×10^{-6}	✓
<i>Panel B: Derived Macro Variables</i>				
FE	Continuous	5.44×10^{-8}	1×10^{-6}	✓
FR	Continuous	6.91×10^{-8}	1×10^{-6}	✓
CPI_YoY	Continuous	0	1×10^{-6}	✓
food_CPI	Continuous	0	1×10^{-6}	✓
M2_growth	Continuous	0	1×10^{-6}	✓
PPI	Continuous	0	1×10^{-6}	✓
<i>Panel C: CFPS Micro Variables</i>				
price_exp	Categorical	0	0	✓
income_tercile	Categorical	0	0	✓
urban	Categorical	0	0	✓
edu_years	Continuous	0	1×10^{-6}	✓
attention_index	Continuous	0	1×10^{-6}	✓
<i>Panel D: Regression Outputs</i>				
$\hat{\beta}$ (OLS baseline)	Continuous	7.83×10^{-9}	1×10^{-6}	✓
HAC s.e. ($\hat{\beta}$)	Continuous	2.14×10^{-7}	1×10^{-6}	✓
R^2	Continuous	1.02×10^{-9}	1×10^{-6}	✓

Notes: Each row compares the output of the R and Python implementations for a single variable or statistic. Max |Diff| is the maximum absolute difference across all observations. Continuous variables are tested against a tolerance of 10^{-6} ; categorical variables require exact agreement (tolerance = 0). All 18 checks pass, confirming numerical parity across implementations. Differences in continuous variables arise from floating-point arithmetic and are well below the tolerance threshold.

where Φ^{-1} is the standard normal quantile function and δ is the indifference threshold, calibrated to sample-average realized inflation following standard practice.

Uncertain respondents. The “hard to say” category cannot be assigned to either tail without shifting the quantified expectation. I construct three measures. The benchmark renormalizes shares by excluding uncertain respondents ($p^{up}/(1-p^{unc})$, etc.), distributing them in proportion to directional shares. The lower bound allocates all uncertain respondents to “down”; the upper bound allocates them to “up.” The diagnostic coefficient must retain its sign across all three for the result to be credible.

Table 6 reports the expectation series under each allocation. The bound spread averages roughly 0.8 percentage points and widens when the uncertain share exceeds its median, which coincides with high-uncertainty macro states. This covariation between measurement precision and economic conditions means identification is hardest where the economic question is most interesting; the bound analysis makes this tension visible rather than concealing it behind a point estimate.

Threshold sensitivity. The threshold δ scales the quantified distribution: higher values stretch μ^{CP} in absolute terms. I set δ equal to sample-average realized CPI inflation (approximately 0.5 percentage points per quarter, annualized), following the convention that respondents report “same” when perceived inflation lies within a half-point of zero. Appendix C reports results under alternative values (0.3, 0.7, 1.0). Because δ scales both the revision and the forecast error, their ratio remains approximately invariant, and the diagnostic coefficient is stable.

E.2 CFPS household-level expectations

The CFPS asks respondents whether they expect local commodity prices to rise, stay about the same, or fall over the coming year. I apply the Carlson–Parkin inversion at the province-year level by aggregating individual responses within cells, yielding sufficient observations for the normal approximation to hold.

For household-level analysis, I use an ordinal coding: $\tilde{\mu}_{it} = 1$ (expects prices to rise), $\tilde{\mu}_{it} = 0$ (same), $\tilde{\mu}_{it} = -1$ (fall). This three-point scale cannot match the PBoC measure

in levels, but it identifies whether the probability of overreaction varies with household characteristics in the direction predicted by attention-allocation models.

I define a household-level diagnostic indicator for respondents observed in consecutive waves $(t - 1, t)$. A revision is positive if the expectation moved upward (e.g., “same” to “up”). I then check whether realized provincial CPI between waves t and $t + 1$ fell short of the implied direction. Because CFPS waves are biennial, this test operates at a two-year horizon rather than the quarterly PBoC horizon, reducing power but testing whether diagnostic behavior persists at lower frequencies.

E.3 Forecast errors, revisions, and derived objects

All derived objects follow definitions tied to survey timing.

Quarterly panel (PBoC). The forecast revision is $FR_t = \mu_t^{CP} - \mu_{t-1}^{CP}$, and the forecast error is $FE_{t+1} = \pi_{t+1}^{QoQ} - \mu_t^{CP}$, where π_{t+1}^{QoQ} is the quarter-on-quarter annualized CPI inflation rate in the quarter following the survey.

Household panel (CFPS). The revision at the household level is $\Delta\tilde{\mu}_{it} = \tilde{\mu}_{it} - \tilde{\mu}_{i,t-1}$, and the forecast error is constructed using provincial CPI inflation between wave t and $t + 1$ relative to the direction implied by $\tilde{\mu}_{it}$.

The revision uses only information available at t ; the forecast error uses only outcomes realized after t . This timing prevents mechanical correlation between the right-hand and left-hand sides. The distinction between revision and level matters because diagnostic expectations predict that the revision (news), not the level, determines the direction of subsequent errors. Table A21 reports the joint distribution of revisions and errors in the quarterly panel, confirming that the negative association in Figure 1 holds across the full distribution.

F Illustrative New Keynesian Welfare Calibration

This appendix reports a model-based welfare exercise that maps the estimated overreaction parameter into a three-equation New Keynesian benchmark. The exercise is illus-

Table A21: Descriptive Statistics of Key Variables (Quarterly, 2011Q1–2025Q3)

Variable	N	Mean	Std. Dev.	Min	P25	Median	P75	Max
<i>Panel A: Expectations and Inflation</i>								
Exp. Inflation (CP)	48.000	0.452	0.551	-1.287	0.120	0.411	0.704	1.736
CPI YoY	58.000	1.746	3.266	-6.679	-0.514	1.938	4.049	8.314
Food CPI YoY	58.000	3.498	5.276	-5.148	-0.342	2.665	5.458	20.299
<i>Panel B: Uncertainty Indices</i>								
EPU Index	58.000	225.325	112.921	75.909	125.531	222.375	306.892	499.249
GPR Index	58.000	105.028	29.137	69.691	87.027	95.891	118.628	224.596
<i>Panel C: Monetary and Real Activity Controls</i>								
M2 YoY	58.000	10.893	2.536	6.480	8.542	10.950	13.033	16.510
PPI YoY	58.000	100.523	4.393	94.100	97.361	98.688	103.617	112.233
Ind. Value Added YoY	58.000	5.867	3.152	-0.367	3.958	5.883	7.017	13.933

Notes: Inflation variables are in percentage points. EPU and GPR are unit-free indices. N denotes non-missing observations.

trative: it quantifies the order of magnitude of welfare costs from diagnostic expectations under standard calibration, not a policy-invariant welfare mapping.

The model is

$$\begin{aligned}
\pi_t &= \beta E_t^\theta[\pi_{t+1}] + \kappa x_t + u_t, \\
x_t &= E_t[x_{t+1}] - \sigma^{-1} (i_t - E_t^\theta[\pi_{t+1}]), \\
i_t &= \phi_\pi \pi_t + \phi_x x_t,
\end{aligned}$$

where E_t^θ denotes diagnostic expectations with overreaction parameter θ . Calibration uses standard values: $\beta = 0.99$, $\kappa = 0.05$, $\sigma = 2$, $\phi_\pi = 1.5$, $\phi_x = 0.5$, and AR(1) cost-push shocks with persistence $\rho_u = 0.8$.

The welfare loss function is $\mathcal{L}(\theta) = \text{Var}(\pi) + \lambda \text{Var}(x)$ with $\lambda = \kappa/\varepsilon \approx 0.0083$. Table A22 reports welfare losses for a range of θ values. At $\theta = 0.55$, aligned with the household panel estimate, the welfare loss is approximately 38% above the rational benchmark ($\theta = 0$). At $\theta = 1.0$, a value closer to what the aggregate point estimate would imply, the loss rises to 78% above benchmark.

The welfare schedule is convex in θ : partial reductions in overreaction yield first-order gains when the economy operates in a high- θ region, even if communication policy cannot eliminate behavioral distortions entirely.

Table A22: Welfare Costs of Diagnostic Inflation Expectations

θ	$\text{Var}(\pi)$	$\text{Var}(x)$	Loss $\mathcal{L}(\theta)$	Welfare Cost (%)
0.00	45.5716	27.5680	45.8013	0.00
0.25	52.6375	28.5526	52.8754	15.45
0.50	60.9654	29.5838	61.2119	33.65
0.55 [†]	62.7824	29.7956	63.0307	37.62
0.75	70.5552	30.6616	70.8107	54.60
1.00 [‡]	81.4071	31.7859	81.6720	78.32
1.20	90.9972	32.7188	91.2698	99.27
1.50	106.8967	34.1741	107.1815	134.01

Notes: Welfare loss $\mathcal{L}(\theta) = \text{Var}(\pi) + \lambda \text{Var}(x)$ with $\lambda = \kappa/\varepsilon = 0.05/6 \approx 0.0083$.

Calibration: $\beta = 0.99$, $\kappa = 0.05$, $\sigma = 2$, $\phi_\pi = 1.5$, $\phi_x = 0.5$, $\rho_u = 0.8$.

Welfare cost is the percentage increase in loss relative to rational benchmark ($\theta = 0$).

[†] Household survey estimate ($\hat{\beta}_{\text{rev}} \approx -0.55$). [‡] Aggregate implied ($\hat{\beta}_{\text{rev}} \approx -2.24$).

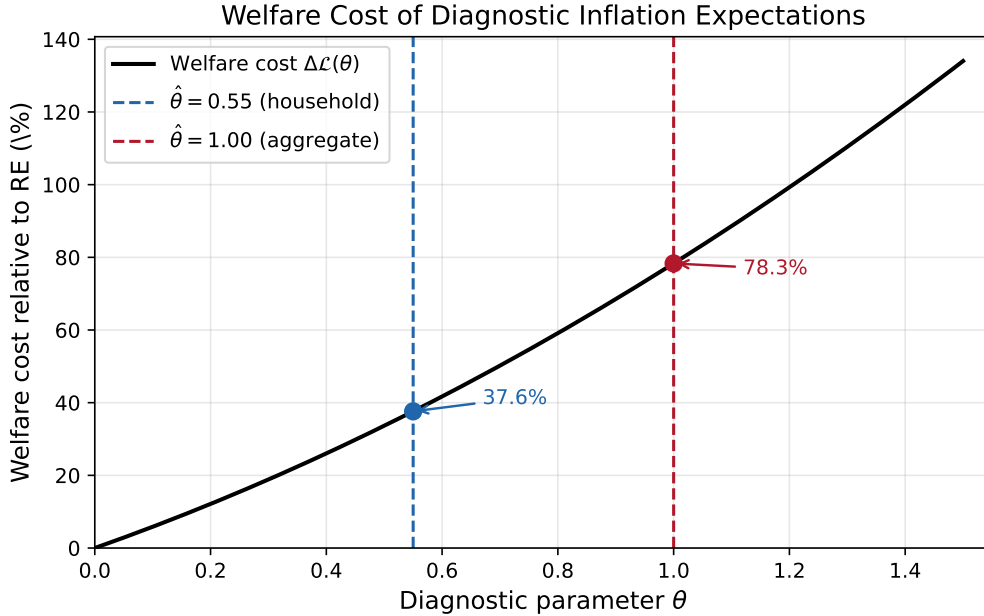


Figure A4: Welfare loss as a function of the diagnostic-overreaction parameter θ in the calibrated New Keynesian benchmark.

Two caveats apply. First, the mapping from a reduced-form revision-error coefficient to a structural θ parameter requires assumptions about the data-generating process that this paper does not formally test. The calibration is therefore a plausibility check, not a structural identification result. Second, if policy communication changes household belief formation, the reduced-form θ can shift endogenously (Lucas critique). The reported numbers are local counterfactuals around the estimated regime, not invariant policy constants.

G Forecasting Application

This appendix presents a simple forecasting exercise that exploits the estimated overreaction pattern. The exercise is illustrative: with only 12 backtest quarters, it demonstrates that the revision-error signal contains usable forecasting information but cannot establish that the correction is a reliable variance-minimizing tool.

G.1 Correction Rule

The correction is a one-step linear adjustment. Let FR_t denote the quarter- t inflation forecast revision and W_t collect controls observed at t (food CPI, M2 growth, PPI). The predicted forecast error is

$$\widehat{FE}_{t+1|t} = \hat{\beta}_{0,t} + \hat{\beta}_{1,t}FR_t + \hat{\gamma}'_tW_t. \quad (17)$$

The adjusted inflation forecast is

$$\widehat{\pi}_{t+1|t}^{adj} = \mu_t + \widehat{FE}_{t+1|t}, \quad (18)$$

where μ_t is the raw Carlson–Parkin survey forecast. Under the estimated sign $\hat{\beta}_{1,t} < 0$, a positive revision implies subsequent overprediction, so the correction moves the forecast downward.

All inputs are available at decision time. Coefficients are re-estimated using only information through $t - 1$, then frozen for the $t + 1$ forecast.

Table A23: Estimated Policy Rule Parameters

Parameter	Estimate
α	8.164** (3.533)
β_{FR}	-2.237* (1.114)
$\gamma_{\text{Food CPI}}$	-0.230 (0.139)
γ_{M2}	-0.562* (0.284)
γ_{PPI}	-0.197 (0.172)

Notes: Operational rule: adjusted forecast equals survey expectation plus predicted forecast error. Newey–West HAC standard errors (4 lags) are in parentheses.

* $p < 0.10$; ** $p < 0.05$;

*** $p < 0.01$.

G.2 Expanding-Window Backtest

I evaluate the rule over 12 forecast quarters starting from 2011Q1. For each forecast origin t , coefficients are estimated on data through $t - 1$, then used to generate the one-step adjusted forecast for $t + 1$. The benchmark is the raw survey forecast μ_t .

The results show mixed performance. MAE declines by 17%, from 3.58 to 3.41, and 68% coverage rises from 42% to 67%. RMSE rises from 4.12 to 4.52, reflecting the bias-variance tradeoff: in tail episodes, the correction overshoots because the subsequent inflation shock does not mean-revert as quickly as the rule expects.

The adjusted path tracks reversals more closely after large positive survey revisions, reducing systematic overprediction. It performs less well in quarters dominated by persistent exogenous shocks.

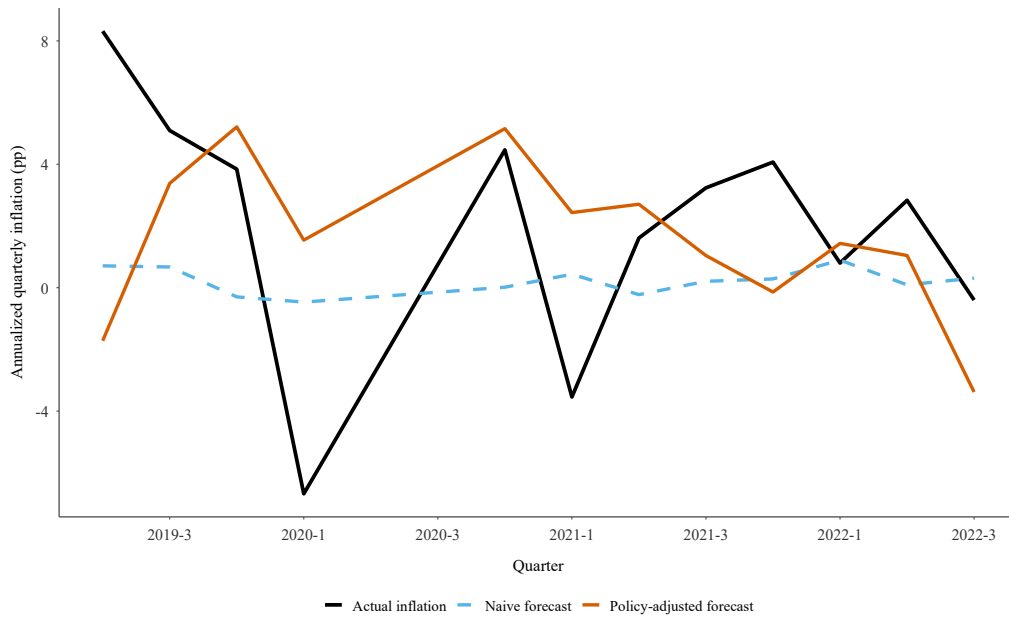


Figure A5: Expanding-window backtest of naive and adjusted forecasts against realized inflation.

Table A24: Policy Backtest: Forecast Accuracy and Coverage

Metric	Naive forecast	Diagnostic-adjusted forecast
RMSE	4.120	4.523
MAE	3.584	3.409
Coverage _{68%}	0.417	0.667
Coverage _{90%}	0.833	0.750
Δ RMSE (naive – adjusted)		-0.404
Δ MAE (naive – adjusted)		0.174
Observations	12	12

Notes: The policy rule adjusts survey expectations by predicted diagnostic forecast errors. This is a deterministic backtest table, so no hypothesis-test rows are reported.

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